Feature-Enhanced Speckle Reduction via Low-Rank and Space-Angle Continuity for Circular SAR Target Recognition

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***Abstract*— With the development of synthetic aperture radar (SAR) system, automatic target recognition (ATR) has attracted wide attention in many decision-making tasks, in which an enhanced feature of SAR image is a powerful tool to improve the recognition accuracy. However, the presence of speckle noise and natural clutter inevitably contaminates SAR images and, thus, degrades image features. In this article, we explicitly address the speckle reduction problem for the circular SAR system, in which the motion of aircraft platform causes continuous angular variations so that different SAR images can be captured with the high interrelationship. By exploiting the underlying low-rank and continuous properties among different SAR images, a method called the ℓ_p-regularized low-rank and space-angle continuity extraction (ℓ_p-LSCE) is proposed to suppress the noise and enhance the target feature. Taking into account the interrelationship between SAR images, we arrange the images in a 3-D tensor to investigate the space-angle continuity of the targets. Furthermore, we develop a robust ℓ_p-regularized scheme to incorporate the low-rank property of targets. Then, the joint optimization problem is solved via the framework of augmented Lagrange multiplier (ALM) with efficient computation of each ALM subproblem. The experimental results of circular SAR data sets of the moving and stationary target acquisition and recognition (MSTAR) and the VideoSAR demonstrate that the proposed method can efficiently despeckle SAR images with well-preserved target features, which is conducive to the improvement of ATR performance.**

***Index Terms*— Automatic target recognition (ATR), circular synthetic aperture radar (SAR), feature enhancement, ℓ_p-regularization, low-rank, space-angle continuity, speckle reduction.**

I. Introduction

AUTOMATIC target recognition (ATR) for synthetic aperture radar (SAR) system has been extensively leveraged in many remote sensing image applications [1]–[12]. Among techniques for ATR, the feature-based methods, which extract features from SAR images to feed into the classifier for recognition [6]–[12], have drawn wide attention and show reasonable performance.

To characterize the visualized feature, the intensity of SAR image in the amplitude domain can be directly used to provide a distinct target structure, which is a common strategy in many ATR algorithms [1]–[5]. In this strategy, the performance of corresponding algorithms is strongly associated with the quality of the images. In real world scenarios, however, the presence of noise and clutter inevitably degrades the quality of image. The speckle noise, generated from the coherent processing of signals in SAR imaging [13], [14], weakens the detailed structure of region of interest such as target and shadow. In addition, SAR images are usually contaminated by natural-clutter false alarms due to non-ideal imaging environments [15]. Because of the interference, the intensity of primary image is not an appropriate feature for the recognition task. How to extract the distinct structure of target while removing the interference of speckle and clutter false alarm is the key component to improve the SAR ATR performance.

In order to eliminate the speckle, many filter-based methods, such as the Lee filter [14], Kuan filter [16], and Frost filter [17], have been proposed for adaptive smoothing of SAR images. In recent years, the diffusion and nonlocal filter-based methods, such as the speckle-reducing anisotropic diffusion (SRAD) [18] and the SAR block-matching 3-D (SAR-BM3D) [19], further improve the despeckling performance in terms of smoothing homogeneous background while preserving edge and texture of target. In addition to SAR-BM3D, the probabilistic patch-based (PPB) method also employs the nonlocal filtering strategy, which is connected with the weighted maximum likelihood estimator to despeckle SAR images in the spatial domain. Interested readers are referred to [21]–[23] and references therein for a general review about speckle reduction methods, which are capable of enhancing the features of SAR images for ATR. It means that feeding the despeckled image rather than the primary image into the classifier contributes to the improvement of recognition accuracy. Nevertheless, most existing methods of speckle reduction are implemented image by image, i.e., they only utilize the information of one single image

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for denoising and, hence, ignore the underlying information between images. When the SAR system works in the circular mode [24], it is capable of forming highly similar images, and hence, the interrelationship between images can provide very useful information to exploit their generic target features for the noise suppression. Different from the abovementioned single-image denoising methods, this article proposes a multiview denoising method, which only works for the circular SAR.

In the circular SAR mode, images of the same target may look quite similar when their illuminated poses of aspect angles only rotate in a small range [24]. In [11], the multiview denoising method is proposed by combining samples of adjacent aspect angles. Since the targets in adjacent samples are strongly related and reside in low-dimensional subspaces, target features are separated from the sparse noise via the robust principal component analysis (RPCA) [25], [26]. The RPCA models apply the ℓ_1-regularized scheme to restrain distinct outliers. However, because of the sparse nature of the ℓ_1-norm [27], [28], the noise suppression via the RPCA tends to be insufficient due to widespread speckles and natural clutters in SAR images. Also, it is observed that the fixed ℓ_1-regularization hardly adapts to the variability of noise caused by the environmental disturbance. By the low-dimensional representation, the low-rank matrix approximation is also applied in [12] to extract the feature. It depicts the manifold structure from noisy SAR images, but the extracted features are not for visualization.

Motivated by the above analysis, in this article, we explicitly address the visualized feature enhancement by despeckling contaminated SAR images, which ensures a reasonable interpretability to guide the subsequent recognition task. Since the scattering mechanism of SAR implies that the image formation within 5° aspect angles is usually stable [29], SAR images captured from the same pose space are highly interrelated. We arrange them to form a 3-D tensor to investigate the intrinsic target structure. For the arranged 3-D tensor, we take the advantage of the low-rank property of target and also observe that the speckle noise and clutter are usually randomly distributed [14]. To enhance the noise robustness, we remodel the RPCA by using an unified ℓ_p-regularized framework with a flexible choice of $1 \leq p \leq 2$, which is conducive to extracting a cleaner target structure while edge features of target are also preserved well.

Apart from the low-rank property, we further characterize the feature of target by its spatial structure. Note that the speckle noise is usually isolated in SAR images with the spatial discontinuity, which is different from the target that tends to occupy a continuous region with smooth intensity values. Hence, the continuity constraint can be enforced to discriminate the target component from the noise component. Furthermore, for the arranged 3-D tensor, the continuous property is extended from the spatial domain to the additional aspect angle domain. Then, we uniformly form the space-angle continuity for the target structure. The constraint of space-angle continuity is efficiently encoded by the 3-D total variation (TV) [30], which shows a reasonable denoising performance.

Then, we jointly exploit the low-rank and continuous property of the target to propose an algorithm, called the ℓ_p-regularized low-rank and space-angle continuity extraction (ℓ_p-LSCE), to enhance the target structure while suppressing the interference caused by speckle noise and background clutter. Our main contributions are summarized as follows.

1) We take the advantage of the interrelationship between SAR images. By arranging images from the same pose space to form a 3-D tensor, we perform the speckle reduction in a tensor manner with characteristics that do not exist in single-image processing.
2) We incorporate the low-rank and continuous properties to enhance the target. More importantly, the enforcement of continuity is extended from the spatial domain to the unified space-angle domain.
3) We remodel the constraint on noise by an ℓ_p-regularization ($1 \leq p \leq 2$), which is experimentally demonstrated to extract the noise-resistant and edge-preserved target structure. Also, the flexible adjustment of p helps to handle varied scenarios.
4) The resultant optimization problem of ℓ_p-LSCE is efficiently solved by the framework of augmented Lagrange multiplier (ALM) [31]. Alternately estimating the target component and noise component, the ALM obtains the optimal solution with fast convergence.

The rest of this article is organized as follows. The problem formulation is introduced in Section II, where the joint enforcement of low-rank and continuity constraint is described in detail. The proposed ℓ_p-LSCE algorithm is presented in Section III, whose optimization is rigorously derived via ALM. The experimental results are shown in Section IV to demonstrate the efficiency, accuracy, and noise robustness of our proposed algorithm. Finally, conclusions are discussed in Section V.

Notations: The $(\cdot)^T$ and $\langle \cdot, \cdot \rangle$ stand for the transpose and inner product. A scalar, a vector, a matrix, and a tensor are denoted by a regular lowercase letter f, a bold lowercase letter $\boldsymbol{f}$, a bold uppercase letter $\boldsymbol{F}$, and a bold calligraphic letter $\boldsymbol{\mathcal{F}}$, respectively. In detail, $[\boldsymbol{f}]_i$ represents the ith entry of the vector $\boldsymbol{f}$, $[\boldsymbol{F}]_{i,j}$ stands for the (i, j)th entry of the matrix $\boldsymbol{F}$, and $[\boldsymbol{\mathcal{F}}]_{x,y,a}$ denotes the (x, y, a)th entry of the three-order tensor $\boldsymbol{\mathcal{F}}$. $\mathbf{vec}(\cdot)$ stands for the vectorization operator, i.e., $\mathbf{vec}(\boldsymbol{F})$ is the vectorization of matrix $\boldsymbol{F}$ while $\mathbf{vec}(\boldsymbol{\mathcal{F}})$ is that of tensor $\boldsymbol{\mathcal{F}}$. Correspondingly, $\mathbf{mat}(\cdot)$ stands for the matrixing operator. Other notations will be introduced in the following sections when required.

II. Problem Formulation

A. Low-Rank Property and RPCA

Target images captured by the circular SAR vary continuously with aspect angles changing from 0° to 360°, and part of which is shown in Fig. 1. When SAR platform illuminates the region of interest from very different aspect angles, there are significant differences between target images since their radar cross sections (RCSs) vary dramatically [11]. Nevertheless, targets within the same class can look quite similar and maintain an intrinsical structure, especially when

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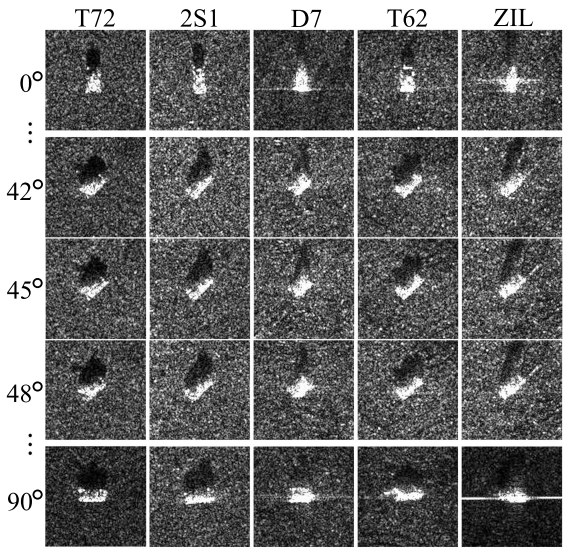


Fig. 1. SAR images of types T72, 2S1, D7, T62, and ZIL131 with varied aspect angles.

their illuminated poses of aspect angle only rotate in a small range. Because of the scattering mechanism of SAR, the pose space divided with 5° increments of aspect angle is usually stable [29]. Therefore, target images captured from the same pose space reside in a low-dimensional subspace and have a natural low-rank property [25].

Apart from targets, SAR images also include the shadow of target and background. The shadow, considered as an auxiliary label for target [32], can be leveraged for the subsequent ATR [1]–[12]. However, the background, where the natural clutter and speckle exist, significantly contaminates SAR images and, hence, degrades the performance of recognition. Actually, speckle noise also weakens the features in region of interest including the target and shadow, as illustrated in Fig. 1. Note that clutters and speckle noises are independently distributed [14] in SAR images and, thus, different from the target structure embedded in a low-dimensional subspace. As a consequence, there is a possibility to obtain the noise-resistant image by extracting the low-rank structure of target.

With regard to each class of targets, their illuminated poses vary over 360° aspect angles, which could be divided into different pose spaces with a specific angular increment (see Section IV-E for the selection of angular increment). Within a certain pose space, suppose that there are a total of d images, each of which has the size of m rows and n columns. We arrange all images in ascending order of aspect angle so that the arranged images can be viewed as a 3-D tensor $\boldsymbol{\mathcal{Y}} \in \mathbb{R}^{m\times n\times d}$. After that, we transform the tensor to the matrix $\boldsymbol{Y} = \mathbf{mat}(\boldsymbol{\mathcal{Y}}) \in \mathbb{R}^{mn\times d}$ by vectorizing each image into a vector and stacking the vector column by column into a matrix. Of particular note is that columns of matrix $\boldsymbol{Y}$ result in a low-rank subspace since the target images in the same pose space are highly interrelated, whereas the statistically independent noise maintains a sparse property. According to the discrepancy between low-rank target and sparse noise, it is logical that we can distinguish those two components in SAR images, respectively. Moreover, a natural image can be regarded as a superposition of the target component and the noise component; thus, the target feature enhancement and noise suppression can be achieved by separating and maximizing the discrepancy between the two. Based on the above analysis, the feature-enhanced speckle reduction problem can be treated by decomposing SAR images into the target component $\boldsymbol{F} \in \mathbb{R}^{mn\times d}$ and the noise component $\boldsymbol{N} \in \mathbb{R}^{mn\times d}$ in the additive format as follows:

$$\begin{aligned} &\min_{\boldsymbol{F},\boldsymbol{N}} \ \operatorname{rank}(\boldsymbol{F}) + \lambda\|\boldsymbol{N}\|_0 \\ &\text{s.t. } \boldsymbol{F} + \boldsymbol{N} = \boldsymbol{Y} \end{aligned} \tag{1}$$

where $\|\boldsymbol{N}\|_0$ denotes the ℓ_0-norm of $\boldsymbol{N}$ that counts the number of nonzero entries, λ represents the regularization parameter to trade off the rank of $\boldsymbol{F}$ and the sparsity of $\boldsymbol{N}$. However, (1) is known as an NP-hard problem due to its highly nonconvexity [26]. To realize a tractable convex relaxation, the nuclear norm is proven to be the convex surrogate of $\operatorname{rank}(\boldsymbol{F})$ [33]. Also, the ℓ_1-norm can be used to approximate the ℓ_0-norm, which leads to the well-known RPCA model [25], [26] as follows:

$$\begin{aligned} &\min_{\boldsymbol{F},\boldsymbol{N}} \|\boldsymbol{F}\|_* + \lambda\|\boldsymbol{N}\|_1 \\ &\text{s.t. } \boldsymbol{F} + \boldsymbol{N} = \boldsymbol{Y} \end{aligned} \tag{2}$$

where $\|\boldsymbol{F}\|_* = \sum_i \sigma_i(\boldsymbol{F})$ represents the nuclear norm of $\boldsymbol{X}$ that equals the sum of all singular values.

In conventional RPCA methods, the regularized scheme for noise matrix $\boldsymbol{N}$ is fixed by the ℓ_1-norm. However, the ℓ_1-regularization hardly adapts to the variability of disturbance in SAR systems. Especially for challenging scenarios where speckle noise and background clutter are widespread, the sparse ℓ_1-regularized scheme cannot suppress their interference sufficiently. In this case, the residual noise of $\boldsymbol{N}$ is naturally embedded into the extracted target structure $\boldsymbol{F}$, which may increase the probability of false alarm and degrade the subsequent ATR performance.

To realize a more flexible and noise-robust regularization, we remodel the RPCA by using the unified ℓ_p-regularized framework given by

$$\begin{aligned} &\min_{\boldsymbol{F},\boldsymbol{N}} \|\boldsymbol{F}\|_* + \lambda\|\boldsymbol{N}\|_p^p \\ &\text{s.t. } \boldsymbol{F} + \boldsymbol{N} = \boldsymbol{Y} \end{aligned} \tag{3}$$

where the ℓ_p-norm of matrix $\boldsymbol{N}$ is defined as follows:

$$\|\boldsymbol{N}\|_p = \left(\sum_{(i,j)} |[\boldsymbol{N}]_{i,j}|^p\right)^{1/p}. \tag{4}$$

It is worth pointing out that $1 \le p \le 2$ is adopted in the proposed framework to ensure the convexity, where the case of $p = 2$ is consistent with the Frobenius norm and $p = 1$ reduces to the common-used ℓ_1-norm in the conventional RPCA. Under the unified ℓ_p-regularization, a smaller value of p yields a sparser solution of $\boldsymbol{N}$. Compared with the choice of $p = 1$, using $p = 2$ would produce a more clean target structure in $\boldsymbol{F}$ by capturing denser noise component in $\boldsymbol{N}$, but at the cost of oversmoothed features such as edges, textures, and point scatterers. Hence, a tradeoff strategy of $1 < p < 2$

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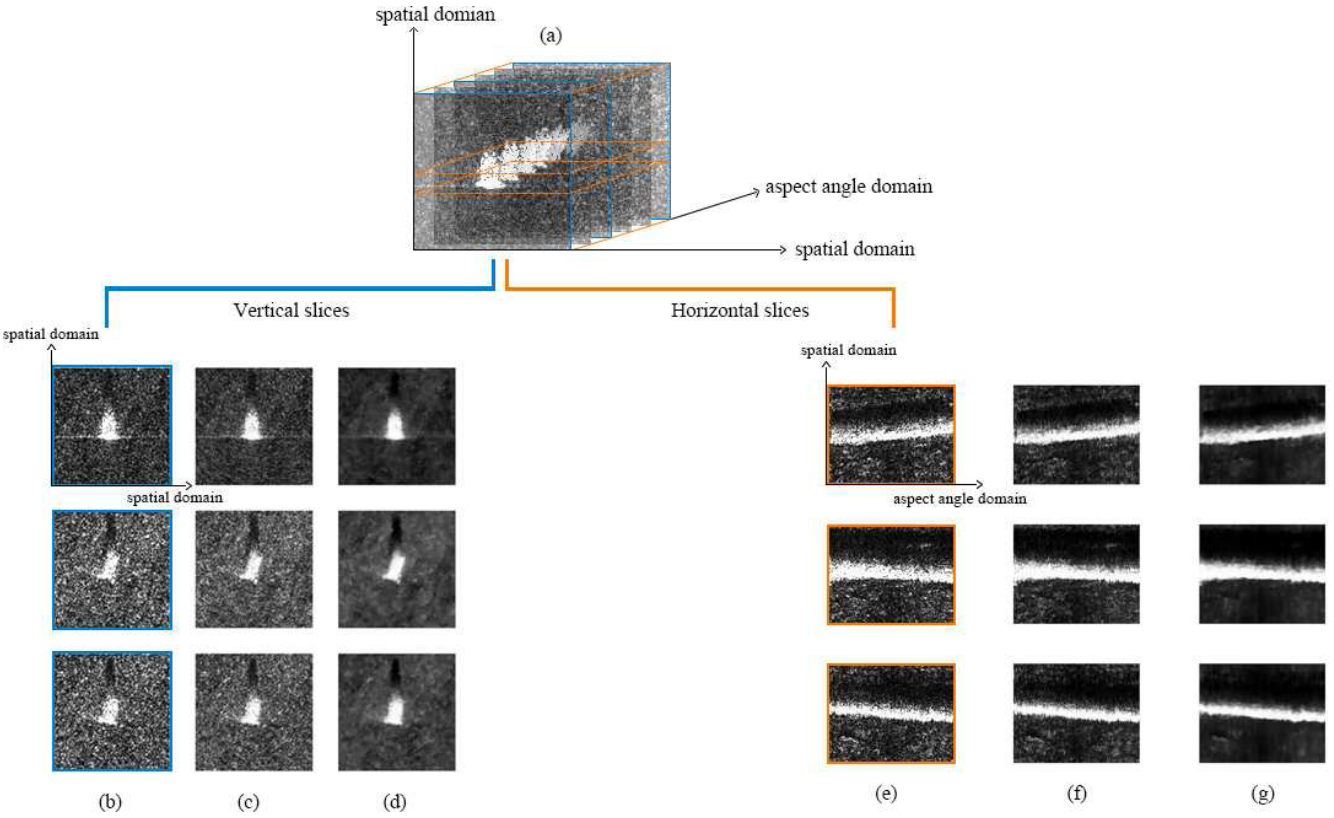


Fig. 2. Illustration of the space-angle continuity for the target structure. We arrange SAR images captured from similar aspect angles to construct a 3-D tensor of (a). Then, we cut the tensor vertically and horizontally into the 2-D slices. Vertical slices of (b) are original SAR images, which support the spatial continuity of the target. By using our proposed algorithm with and without the spatial continuity constraint (setting $[w_x, w_y, w_a] = [1, 1, 1]$ for the constraint case while $[w_x, w_y, w_a] = [0, 0, 1]$ for the unconstraint case), (c) and (d) show the despeckling results of vertical slices, respectively. Correspondingly, horizontal slices of (e) validate that the target is continuous along the aspect angle axis and, thus, support the aspect angle continuity. With and without the angle continuity constraint by setting $[w_x, w_y, w_a] = [1, 1, 1]$ and $[w_x, w_y, w_a] = [1, 1, 0]$, (f) and (g) show the despeckling results of horizontal slices, respectively. We see that without the continuity constraint in the spatial or aspect angle domain, the resulting slice images are contaminated by the noise. In contrast, by the unified enforcement of space-angle continuity, our method can not only preserve the target structure precisely but also filter out the noise sufficiently.

is of particular interest, which is conducive to sufficient noise suppression while the detailed features of targets are also well-preserved. In this article, we flexibly adjust the value of p to obtain a reasonable result of target structure (see Section IV).

B. Space-Angle Continuity and TV

From the ATR perspective, we aim at enhancing a pure target structure with noise and background clutter being suppressed. Apart from the low-rank property, the target structure can be further characterized in the spatial domain of image. As shown in Fig. 1, the target, including its shadow, tends to occupy a certain continuous region with smooth intensity values in SAR image, whereas the noise is usually isolated and shows a discontinuous property. Therefore, the spatial continuity constraint can be introduced into our model to enhance the underlying target structure.

Note that 2-D SAR images with similar aspect angles are arranged into the 3-D tensor in our model; thus, we investigate whether the continuity constraint can be extended from two spatial domains to the third aspect angle domain. Here, we cut the tensor vertically and horizontally into the 2-D slices, as shown in Fig. 2. Vertical slices are original SAR images that support the continuity in two spatial domains. Correspondingly, horizontal slices are based on the spatial domain versus aspect angle domain, along which the occupied regions of targets are continuous with a stripe-like distribution. In these stripe-like regions, as shown in Fig. 2(e), the intensities of targets are salient and have the property of continuity, which is verified to exist not only in the spatial domain but also in the aspect angle domain. In this article, we uniformly call the property of target structure as the space-angle continuity. The difference between applying only spatial continuity constraint and unified space-angle continuity constraint is illustrated in Fig. 2(f) and (g). Without the angle continuity constraint, Fig. 2(f) shows the target enhancement result that cannot filter out the speckle noise sufficiently. By enforcing the unified space-angle constraint, the proposed method achieves the better enhancement of angle continuity in Fig. 2(g), which shows a superior noise robustness while precisely preserving the target features.

To quantize the degree of continuity, the first-order forward difference is used. For a tensor $\boldsymbol{\mathcal{F}}$, the difference operators in the spatial domain (∇_x, ∇_y) and the aspect angle domain ∇_a are defined as follows, respectively:

$$\nabla_x \boldsymbol{\mathcal{F}} := [\boldsymbol{\mathcal{F}}]_{x+1,y,a} - [\boldsymbol{\mathcal{F}}]_{x,y,a} \tag{5}$$

$$\nabla_y \boldsymbol{\mathcal{F}} := [\boldsymbol{\mathcal{F}}]_{x,y+1,a} - [\boldsymbol{\mathcal{F}}]_{x,y,a} \tag{6}$$

$$\nabla_a \boldsymbol{\mathcal{F}} := [\boldsymbol{\mathcal{F}}]_{x,y,a+1} - [\boldsymbol{\mathcal{F}}]_{x,y,a}. \tag{7}$$

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For the vector $\boldsymbol{f} = \mathbf{vec}(\boldsymbol{\mathcal{F}}) \in \mathbb{R}^{mnd\times 1}$, the difference operation in x domain (analogous definitions in y and a domains) is given by $\boldsymbol{D}_x\boldsymbol{f} = \mathbf{vec}(\nabla_x\boldsymbol{\mathcal{F}})$, where $\boldsymbol{D}_x \in \mathbb{R}^{mnd\times mnd}$. Then, the concatenated difference operation in the unified space-angle domains is denoted by $\boldsymbol{Df}$, where $\boldsymbol{D} = [\boldsymbol{D}_x^T\ \boldsymbol{D}_y^T\ \boldsymbol{D}_a^T]^T \in \mathbb{R}^{3mnd\times mnd}$.

Then, we propose to employ the 3-D TV to efficiently encode the constraint of space-angle continuity. For the flexible control of each domain, we introduce three scalar weights w_x, w_y, w_a to define the TV-norm as follows:

$$\|\boldsymbol{F}\|_{\mathrm{TV}} = \sum_i \sqrt{w_x^2[\boldsymbol{D}_x\boldsymbol{f}]_i^2 + w_y^2[\boldsymbol{D}_y\boldsymbol{f}]_i^2 + w_a^2[\boldsymbol{D}_a\boldsymbol{f}]_i^2}. \quad (8)$$

Equation (8) induces the TV-norm in an isotropic form [30] that can be rewritten by the $\ell_{2,1}$-norm [34]

$$\|\boldsymbol{F}\|_{\mathrm{TV}} = \|\mathbf{mat}(\boldsymbol{Df})\|_{2,1} \quad (9)$$

where $\mathbf{mat}(\boldsymbol{Df}) = [w_x\boldsymbol{D}_x\boldsymbol{f}\ w_y\boldsymbol{D}_y\boldsymbol{f}\ w_a\boldsymbol{D}_a\boldsymbol{f}] \in \mathbb{R}^{mnd\times 3}$. Correspondingly, the anisotropic form [30] induces a substitute TV-norm as follows:

$$\|\boldsymbol{F}\|_{\mathrm{TV}} = \sum_i (w_x|[\boldsymbol{D}_x\boldsymbol{f}]_i| + w_y|[\boldsymbol{D}_y\boldsymbol{f}]_i| + w_a|[\boldsymbol{D}_a\boldsymbol{f}]_i|) \quad (10)$$

which is equivalent to the ℓ_1-norm of $\mathbf{mat}(\boldsymbol{Df})$. The anisotropic TV enforces the continuity with a nature of sparsity while the isotropic TV can provide more dense and salient suppression to discontinuous changes generally regarded as noises. In this article, we utilize the isotropic TV to encode the space-angle continuity, which is experimentally demonstrated to have better performance in terms of noise resistance for the ATR problem.

III. ℓ_p-LSCE

By incorporating the low-rank and continuity knowledge of target structure in the space-angle domain, we propose a feature-oriented algorithm, called the ℓ_p-LSCE, to enhance the target from the noisy environment, which yields the following optimization problem:

$$\min_{\boldsymbol{F},\boldsymbol{N}} \|\boldsymbol{F}\|_* + \lambda_1\|\boldsymbol{F}\|_{\mathrm{TV}} + \lambda_2\|\boldsymbol{N}\|_p^p \quad \text{s.t. } \boldsymbol{F}+\boldsymbol{N} = \boldsymbol{Y}. \quad (11)$$

It is worth mentioning that the proposed ℓ_p-LSCE algorithm is capable of denoising a generic image rather than just removing the speckle. Our model in (11) relies on the fact that the noise in different SAR images is usually statistically independent, unlike the underlying target structure that lies in a joint low-rank and continuous subspace. Based upon the above discrepancy between the target component $\boldsymbol{F}$ and the noise component $\boldsymbol{N}$, we can extract and distinguish these two components, respectively, in the generic image to achieve denoising.

A. ALM

Equation (11) derives a constrained optimization problem that is challenging to be solved directly. Instead, we apply the framework of ALM [26], [31] to efficiently solve (11), which minimizes the dual-form of original constrained problem by a series of unconstrained subproblems. Within the framework of ALM, we first introduce an auxiliary variable $\boldsymbol{E}$ to reformulate the problem as follows:

$$\min_{\boldsymbol{E},\boldsymbol{F},\boldsymbol{N}} \|\boldsymbol{E}\|_* + \lambda_1\|\boldsymbol{F}\|_{\mathrm{TV}} + \lambda_2\|\boldsymbol{N}\|_p^p \quad \text{s.t. } \boldsymbol{E}+\boldsymbol{N} = \boldsymbol{Y},\quad \boldsymbol{F} = \boldsymbol{E}. \quad (12)$$

Then, the augmented Lagrangian function of (12) is defined as follows:

$$\begin{aligned}\mathcal{L}_\mu(\boldsymbol{E},\boldsymbol{F},\boldsymbol{N},\boldsymbol{O},\boldsymbol{Q}) &= \|\boldsymbol{E}\|_* + \lambda_1\|\boldsymbol{F}\|_{\mathrm{TV}} + \lambda_2\|\boldsymbol{N}\|_p^p \\ &\quad + \langle\boldsymbol{O},\boldsymbol{Y}-\boldsymbol{E}-\boldsymbol{N}\rangle + \frac{\mu}{2}\|\boldsymbol{Y}-\boldsymbol{E}-\boldsymbol{N}\|_F^2 \\ &\quad + \langle\boldsymbol{Q},\boldsymbol{E}-\boldsymbol{F}\rangle + \frac{\mu}{2}\|\boldsymbol{E}-\boldsymbol{F}\|_F^2 \end{aligned} \quad (13)$$

where μ is the positive penalty parameter, and $\boldsymbol{O} \in \mathbb{R}^{mn\times d}$ and $\boldsymbol{Q} \in \mathbb{R}^{mn\times d}$ are the Lagrange multipliers (dual variables) associated with equality constraints in (12).

Because (12) is closed, proper and convex, ALM can approximate the optimal primal and dual variables by finding a saddle point of the augmented Lagrangian [31]. Specifically, the primal variables are iteratively updated as follows:

$$(\boldsymbol{E}_{k+1},\boldsymbol{F}_{k+1},\boldsymbol{N}_{k+1}) = \arg\min_{\boldsymbol{E},\boldsymbol{F},\boldsymbol{N}} \mathcal{L}_\mu(\boldsymbol{E},\boldsymbol{F},\boldsymbol{N},\boldsymbol{O}_k,\boldsymbol{Q}_k). \quad (14)$$

Note that the gradient of the augmented Lagrangian function with respect to the dual-variable $\boldsymbol{O}$ is

$$\frac{\partial\mathcal{L}_\mu(\boldsymbol{E}_{k+1},\boldsymbol{F}_{k+1},\boldsymbol{N}_{k+1},\boldsymbol{O}_k,\boldsymbol{Q}_k)}{\partial\boldsymbol{O}} = \boldsymbol{Y}-\boldsymbol{E}_{k+1}-\boldsymbol{N}_{k+1}. \quad (15)$$

Then, the gradient ascent with the optimal step size of μ_k can be employed to update $\boldsymbol{O}_{k+1}$ as follows:

$$\boldsymbol{O}_{k+1} = \boldsymbol{O}_k + \mu_k(\boldsymbol{Y}-\boldsymbol{E}_{k+1}-\boldsymbol{N}_{k+1}). \quad (16)$$

An analogous procedure derives the dual-variable $\boldsymbol{O}_{k+1}$ as follows:

$$\boldsymbol{Q}_{k+1} = \boldsymbol{Q}_k + \mu_k(\boldsymbol{E}_{k+1}-\boldsymbol{F}_{k+1}). \quad (17)$$

Since the optimal variables in (14) are coupled, their joint optimization is intractable. A strategy to handle the joint optimization problem is the alternating direction minimizing (ADM) strategy [35], [36], which splits (14) into the following three different subproblems involving $\boldsymbol{E}$, $\boldsymbol{F}$, and $\boldsymbol{N}$.

*1) **E-Subproblem**:* Given $\boldsymbol{F}_k$ and $\boldsymbol{N}_k$, we first update $\boldsymbol{E}_{k+1}$ as follows:

$$\begin{aligned}\boldsymbol{E}_{k+1} &= \arg\min_{\boldsymbol{E}} \mathcal{L}_\mu(\boldsymbol{E},\boldsymbol{F}_k,\boldsymbol{N}_k,\boldsymbol{O}_k,\boldsymbol{Q}_k) \\ &= \arg\min_{\boldsymbol{E}} \|\boldsymbol{E}\|_* + \langle\boldsymbol{O}_k,\boldsymbol{Y}-\boldsymbol{E}-\boldsymbol{N}_k\rangle \\ &\quad + \frac{\mu_k}{2}\|\boldsymbol{Y}-\boldsymbol{E}-\boldsymbol{N}_k\|_F^2 + \langle\boldsymbol{Q}_k,\boldsymbol{E}-\boldsymbol{F}_k\rangle \\ &\quad + \frac{\mu_k}{2}\|\boldsymbol{E}-\boldsymbol{F}_k\|_F^2 \\ &= \arg\min_{\boldsymbol{E}} \|\boldsymbol{E}\|_* + \frac{\mu_k}{2}\left(\|\boldsymbol{E}-\boldsymbol{E}_\Lambda\|_F^2 + \|\boldsymbol{E}-\boldsymbol{E}_{\mathrm{V}}\|_F^2\right) \\ &= \arg\min_{\boldsymbol{E}} \|\boldsymbol{E}\|_* + \mu_k\left(\|\boldsymbol{E}-\frac{1}{2}(\boldsymbol{E}_\Lambda+\boldsymbol{E}_{\mathrm{V}})\|_F^2\right) \end{aligned} \quad (18)$$

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where $\boldsymbol{E}_\Lambda = \boldsymbol{Y} - \boldsymbol{N}_k + \mu_k^{-1}\boldsymbol{O}_k$ and $\boldsymbol{E}_\mathrm{V} = \boldsymbol{F}_k - \mu_k^{-1}\boldsymbol{Q}_k$ are used for simplicity. By using the singular value thresholding operator [33], the closed-form solution of (18) is given by

$$\boldsymbol{E}_{k+1} = \boldsymbol{U}\boldsymbol{\mathcal{S}}_{1/(2\mu_k)}(\boldsymbol{\Sigma})\boldsymbol{V}^T \tag{19}$$

where $1/2(\boldsymbol{E}_\Lambda + \boldsymbol{E}_\mathrm{V}) = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ is the singular value decomposition (SVD), and $\boldsymbol{\mathcal{S}}_{1/(2\mu_k)}(\cdot)$ denotes the soft-thresholding operator [37] with the parameter $1/(2\mu_k)$.

*2) **F-Subproblem**:* Next, we update $\boldsymbol{F}_{k+1}$ with the given $\boldsymbol{E}_{k+1}$ and $\boldsymbol{N}_k$ as follows:

$$\begin{aligned}\boldsymbol{F}_{k+1} &= \arg\min_{\boldsymbol{F}} \mathcal{L}_\mu(\boldsymbol{E}_{k+1}, \boldsymbol{F}, \boldsymbol{N}_k, \boldsymbol{O}_k, \boldsymbol{Q}_k)\\ &= \arg\min_{\boldsymbol{F}} \lambda_1\|\boldsymbol{F}\|_\mathrm{TV} + \langle \boldsymbol{Q}_k, \boldsymbol{E}_{k+1} - \boldsymbol{F}\rangle\\ &\quad + \frac{\mu_k}{2}\|\boldsymbol{E}_{k+1} - \boldsymbol{F}\|_F^2\\ &= \arg\min_{\boldsymbol{F}} \lambda_1\|\boldsymbol{F}\|_\mathrm{TV} + \frac{\mu_k}{2}\|\boldsymbol{F} - \boldsymbol{F}_\Lambda\|_F^2 \end{aligned}\tag{20}$$

where $\boldsymbol{F}_\Lambda = \boldsymbol{E}_{k+1} + \mu_k^{-1}\boldsymbol{Q}_k$ is defined. With the definition of TV-norm, we equivalently rewrite (20) as the vector form

$$\min_{\boldsymbol{f}} \lambda_1\|\mathbf{mat}(\boldsymbol{D}\boldsymbol{f})\|_{2,1} + \frac{\mu_k}{2}\|\boldsymbol{f} - \boldsymbol{f}_\Lambda\|_2^2 \tag{21}$$

where $\boldsymbol{f} = \mathbf{vec}(\boldsymbol{F})$ and $\boldsymbol{f}_\Lambda = \mathbf{vec}(\boldsymbol{F}_\Lambda)$. Then, an auxiliary variable $\boldsymbol{g} \in \mathbb{R}^{mnd\times 1}$ is introduced to separate the objective. Accordingly, an additional equality constraint $\boldsymbol{D}\boldsymbol{f} = \boldsymbol{g}$ is required to reformulate (21) as follows:

$$\begin{aligned}&\min_{\boldsymbol{f},\boldsymbol{g}} \lambda_1\|\mathbf{mat}(\boldsymbol{g})\|_{2,1} + \frac{\mu_k}{2}\|\boldsymbol{f} - \boldsymbol{f}_\Lambda\|_2^2\\ &\text{s.t. } \boldsymbol{D}\boldsymbol{f} = \boldsymbol{g}.\end{aligned}\tag{22}$$

Then, the augmented Lagrangian of (22) is defined as follows:

$$\begin{aligned}\mathcal{L}_\eta(\boldsymbol{f}, \boldsymbol{g}, \boldsymbol{z}) = \lambda_1\|\mathbf{mat}(\boldsymbol{g})\|_{2,1} + \frac{\mu_k}{2}\|\boldsymbol{f} - \boldsymbol{f}_\Lambda\|_2^2\\ + \langle \boldsymbol{z}, \boldsymbol{g} - \boldsymbol{D}\boldsymbol{f}\rangle + \frac{\eta}{2}\|\boldsymbol{g} - \boldsymbol{D}\boldsymbol{f}\|_2^2\end{aligned}\tag{23}$$

where η denotes the penalty parameter and $\boldsymbol{z} \in \mathbb{R}^{mnd\times 1}$ is the Lagrange multiplier.

Again, the resultant problem can be optimized with the strategy of ADM over $\boldsymbol{f}$, $\boldsymbol{g}$, and $\boldsymbol{z}$. The iterative procedures are expressed by the following steps.

1) Updating $\boldsymbol{f}_{t+1}$ with the fixed $\boldsymbol{g}_t$ and $\boldsymbol{z}_t$ as follows:

$$\begin{aligned}\boldsymbol{f}_{t+1} &= \arg\min_{\boldsymbol{f}} \mathcal{L}_\eta(\boldsymbol{f}, \boldsymbol{g}_t, \boldsymbol{z}_t)\\ &= \arg\min_{\boldsymbol{f}} \mu_k\|\boldsymbol{f} - \boldsymbol{f}_\Lambda\|_2^2 + \eta_t\left\|\boldsymbol{g}_t - \boldsymbol{D}\boldsymbol{f} + \eta_t^{-1}\boldsymbol{z}_t\right\|_2^2.\end{aligned}\tag{24}$$

Since (24) is quadratic over $\boldsymbol{f}$, the following linear equation gives the optimality:

$$(\mu_k\boldsymbol{I} + \eta_t\boldsymbol{D}^T\boldsymbol{D})\boldsymbol{f} = \mu_k\boldsymbol{f}_\Lambda + \eta_t\boldsymbol{D}^T\boldsymbol{g}_t + \boldsymbol{D}^T\boldsymbol{z}_t. \tag{25}$$

Note that the direct method, which computes the Moore–Penrose pseudoinverse of $(\mu_k\boldsymbol{I} + \eta_t\boldsymbol{D}^T\boldsymbol{D})$, can be used to carry out the $\boldsymbol{f}$-update. However, the computation of the Moore–Penrose pseudoinverse is expensive, especially when the size of $\boldsymbol{D}$ is large. Thanks to the block-circulant structure of $\boldsymbol{D}$, it can be efficiently diagonalized by the 3-D discrete Fourier transform [38]. The exploited structure of $\boldsymbol{D}$ derives the optimal solution as follows:

$$\boldsymbol{f}_{t+1} = \mathbf{ifftn}\left(\frac{\mathbf{fftn}(\mu_k\boldsymbol{f}_\Lambda + \eta_t\boldsymbol{D}^T\boldsymbol{g}_t + \boldsymbol{D}^T\boldsymbol{z}_t)}{\mu_k\boldsymbol{1} + \eta_t\boldsymbol{D}_\Lambda}\right) \tag{26}$$

where $\boldsymbol{D}_\Lambda = w_x^2|\mathbf{fftn}(\boldsymbol{D}_x)|^2 + w_y^2|\mathbf{fftn}(\boldsymbol{D}_y)|^2 + w_a^2|\mathbf{fftn}(\boldsymbol{D}_a)|^2$, $\mathbf{fftn}(\cdot)$ represents the 3-D discrete Fourier transform operator whose inverse transform is denoted by the operator $\mathbf{ifftn}(\cdot)$. The square and division in (26) are performed in the entrywise way. Note that the denominator in (26) remains constant, it can be precomputed outside the iteration to avoid extra overhead.

2) Updating $\boldsymbol{g}_{t+1}$ with the fixed $\boldsymbol{f}_{t+1}$ and $\boldsymbol{z}_t$ as follows:

$$\begin{aligned}\boldsymbol{g}_{t+1} &= \arg\min_{\boldsymbol{g}} \mathcal{L}_\eta(\boldsymbol{g}, \boldsymbol{f}_{t+1}, \boldsymbol{z}_t)\\ &= \arg\min_{\boldsymbol{g}} \lambda_1\|\mathbf{mat}(\boldsymbol{g})\|_{2,1} + \frac{\eta_t}{2}\left\|\boldsymbol{g} - \boldsymbol{D}\boldsymbol{f}_{t+1} + \eta_t^{-1}\boldsymbol{z}_t\right\|_2^2.\end{aligned}\tag{27}$$

We characterize the solution and the Lagrange multiplier by each domain as $\boldsymbol{g}_{t+1} = [\boldsymbol{g}_{t+1,x}^T\ \boldsymbol{g}_{t+1,y}^T\ \boldsymbol{g}_{t+1,a}^T]^T$ and $\boldsymbol{z}_t = [\boldsymbol{z}_{t,x}^T\ \boldsymbol{z}_{t,y}^T\ \boldsymbol{z}_{t,a}^T]^T$, respectively. Then, the solution can be solved with respect to each domain

$$\boldsymbol{g}_{t+1,i} = \max\{\boldsymbol{v} - \lambda_1/\eta_t, 0\} \cdot \frac{\boldsymbol{v}_i}{\boldsymbol{v}}, \quad i \in \{x, y, a\} \tag{28}$$

where $\boldsymbol{v}_i = \boldsymbol{D}_i\boldsymbol{f}_{t+1} - \eta_t^{-1}\boldsymbol{z}_{t,i}$, $\boldsymbol{v} = \max\{(\sum_i|\boldsymbol{v}_i|^2)^{1/2}, \delta\}$ and δ is a small constant to guarantee the stability. The square, division, and dot multiplication in (28) are conducted entrywise, which is analogous to those in (26).

3) Updating $\boldsymbol{z}_{t+1}$ with the fixed $\boldsymbol{f}_{t+1}$ and $\boldsymbol{g}_{t+1}$ as follows:

$$\boldsymbol{z}_{t+1} = \boldsymbol{z}_t + \eta_t(\boldsymbol{g}_{t+1} - \boldsymbol{D}\boldsymbol{f}_{t+1}). \tag{29}$$

Finally, the penalty parameter η_{t+1} can be adaptively updated in the following form [39]:

$$\eta_{t+1} = \begin{cases}\rho\eta_t, & \text{if } \dfrac{||\boldsymbol{g}_{t+1} - \boldsymbol{D}\boldsymbol{f}_{t+1}||_2}{||\boldsymbol{g}_t - \boldsymbol{D}\boldsymbol{f}_t||_2} \geq \varepsilon\\ \eta_t, & \text{otherwise.}\end{cases} \tag{30}$$

Also, the update of the penalty parameter μ_k can similarly follow that of η_t and, hence, is omitted here for succinctness. The entire steps to solve the $\boldsymbol{F}$-subproblem are summarized in Algorithm 1.

*3) **N-Subproblem**:* Given $\boldsymbol{E}_{k+1}$ and $\boldsymbol{F}_{k+1}$, this subproblem is described as follows:

$$\begin{aligned}\boldsymbol{N}_{k+1} &= \arg\min_{\boldsymbol{N}} \mathcal{L}_\mu(\boldsymbol{E}_{k+1}, \boldsymbol{F}_{k+1}, \boldsymbol{N}, \boldsymbol{O}_k, \boldsymbol{Q}_k)\\ &= \arg\min_{\boldsymbol{N}} \lambda_2\|\boldsymbol{N}\|_p^p + \langle \boldsymbol{O}_k, \boldsymbol{Y} - \boldsymbol{E}_{k+1} - \boldsymbol{N}\rangle\\ &\quad + \frac{\mu_k}{2}\|\boldsymbol{Y} - \boldsymbol{E}_{k+1} - \boldsymbol{N}\|_F^2\\ &= \arg\min_{\boldsymbol{N}} \lambda_2\|\boldsymbol{N}\|_p^p + \frac{\mu_k}{2}\|\boldsymbol{N} - \boldsymbol{N}_\Lambda\|_F^2\end{aligned}\tag{31}$$

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where $\boldsymbol{N}_\Lambda = \boldsymbol{Y} - \boldsymbol{E}_{k+1} + \mu_k^{-1}\boldsymbol{O}_k$ is used for simplicity. Apparently, the solution of (31) varies with the choice of p. Here, we distinguish three different cases of p.

1) When $p = 1$, it represents the commonly used ℓ_1-norm, from which it is not difficult to derive the solution as follows:

$$\boldsymbol{N}_{k+1} = \boldsymbol{\mathcal{S}}_{\lambda_2/\mu_k}(\boldsymbol{N}_\Lambda). \tag{32}$$

2) When $1 < p < 2$, the global optimality can be obtained in an entrywise way due to the scalar convexity of (31) but requires an iterative procedure because of the high nonlinearity [40]. The numerical approximation methods can be used to obtain the solution efficiently

$$[\boldsymbol{N}_{k+1}]_{i,j} = \text{sign}([\boldsymbol{N}_\Lambda]_{i,j})[\boldsymbol{N}^*]_{i,j} \tag{33}$$

where $[\boldsymbol{N}^*]_{i,j}$ is the unique root bounded in $[0, |[\boldsymbol{N}_\Lambda]_{i,j}|]$ of the following equation:

$$f([\boldsymbol{N}^*]_{i,j}) = p\lambda_2[\boldsymbol{N}^*]_{i,j}^{p-1} + \mu_k([\boldsymbol{N}^*]_{i,j} - |[\boldsymbol{N}_\Lambda]_{i,j}|) = 0. \tag{34}$$

3) The case of $p = 2$ derives the Frobenius norm optimization problem, which gives the closed-form expression as follows:

$$\boldsymbol{N}_{k+1} = (2\lambda_2 + \mu_k)^{-1}\mu_k\boldsymbol{N}_\Lambda. \tag{35}$$

The complete steps of our proposed algorithm are summarized in Algorithm 2, where the iteration can be viewed as the outer loop. Note that step 3 in each outer loop requires to iteratively update $\boldsymbol{F}$ via Algorithm 1, in which the iteration is considered as the inner loop.

Algorithm 1 Solver of the $\boldsymbol{F}$-Subproblem

Input: $\boldsymbol{F}_\Lambda \in \mathbb{R}^{mn\times d}$, λ_1, μ_k, ρ, and ε.
Initialization: $\boldsymbol{f}_0 = \boldsymbol{g}_0 = \boldsymbol{z}_0 = \boldsymbol{0}$, $w_x = w_y = w_a = 1$, η_0, and compute $\textbf{fftn}(\boldsymbol{D}_x)$, $\textbf{fftn}(\boldsymbol{D}_y)$, $\textbf{fftn}(\boldsymbol{D}_a)$.
for $t = 0, 1, \cdots$ **do**
$\boldsymbol{f}_{t+1} = \textbf{iffftn}\left(\frac{\textbf{fftn}(\mu_k\boldsymbol{f}_\Lambda + \eta_t\boldsymbol{D}^T\boldsymbol{g}_t + \boldsymbol{D}^T\boldsymbol{z}_t)}{\mu_k\boldsymbol{1} + \eta_t\boldsymbol{D}_\Lambda}\right)$
$\boldsymbol{g}_{t+1,i} = \max\{\boldsymbol{v} - \lambda_1/\eta_t, 0\} \cdot \frac{\boldsymbol{v}_i}{\boldsymbol{v}}, \quad i \in \{x, y, a\}.$
$\boldsymbol{z}_{t+1} = \boldsymbol{z}_t + \eta_t(\boldsymbol{g}_{t+1} - \boldsymbol{D}\boldsymbol{f}_{t+1})$
Update η_{t+1} via (30).
Stop if a termination condition is satisfied.
end for
Output: the optimal solution $\boldsymbol{F}_{t+1}$ formed by $\boldsymbol{f}_{t+1}$.

B. Computational Complexity Analysis

Suppose that we are given an input tensor of SAR images $\boldsymbol{\mathcal{Y}} \in \mathbb{R}^{m\times n\times d}$. In each outer loop, the computational cost of the proposed algorithm consists of update procedures over three primal variables $\boldsymbol{E}$, $\boldsymbol{F}$, and $\boldsymbol{N}$ and two dual variables $\boldsymbol{O}$ and $\boldsymbol{Q}$, respectively.

To update $\boldsymbol{E}$ requires to compute the SVD with a complexity of $\mathcal{O}(mnd \cdot \min(mn, d))$ [26], [33]. Then, the thresholding operator yields a shrank singular value matrix of rank r, where $r \ll \min(mn, d)$ is satisfied due to the low-rank property. The solution derives from the multiplication of matrices via (19), which has the complexity $\mathcal{O}((mn + d)r^2)$. Hence, the total computational cost of $\boldsymbol{X}$-subproblem is $\mathcal{O}(mnd \cdot \min(mn, d))$.

Algorithm 2 ℓ_p-LSCE: Optimization of (11) via the ALM Method

Input: $\boldsymbol{\mathcal{Y}} \in \mathbb{R}^{m\times n\times d}$, p, λ_1, λ_2, $\mu_{\max}$, ρ, and ε.
Initialization: $\boldsymbol{E}_0 = \boldsymbol{F}_0 = \boldsymbol{N}_0 = \boldsymbol{O}_0 = \boldsymbol{Q}_0 = \boldsymbol{0}$, and μ_0.
for $k = 0, 1, \cdots$ **do**
$\boldsymbol{E}_{k+1} = \boldsymbol{U}\boldsymbol{\mathcal{S}}_{1/(2\mu_k)}(\boldsymbol{\Sigma})\boldsymbol{V}^T$
Update $\boldsymbol{F}_{k+1}$ via Algorithm 1.
if $p = 1$ **then**
$\boldsymbol{N}_{k+1} = \boldsymbol{\mathcal{S}}_{\lambda_2/\mu_k}(\boldsymbol{N}_\Lambda)$
else if $1 < p < 2$ **then**
$[\boldsymbol{N}_{k+1}]_{i,j} = \text{sign}([\boldsymbol{N}_\Lambda]_{i,j})[\boldsymbol{N}^*]_{i,j}$
else
$\boldsymbol{N}_{k+1} = (2\lambda_2 + \mu_k)^{-1}\mu_k\boldsymbol{N}_\Lambda$
end if
$\boldsymbol{O}_{k+1} = \boldsymbol{O}_k + \mu_k(\boldsymbol{Y} - \boldsymbol{E}_{k+1} - \boldsymbol{N}_{k+1})$
$\boldsymbol{Q}_{k+1} = \boldsymbol{Q}_k + \mu_k(\boldsymbol{E}_{k+1} - \boldsymbol{F}_{k+1})$
Update $\mu_{k+1} = \min(\rho\mu_k, \mu_{\max})$.
Stop if a termination condition is satisfied.
end for
Output: the optimal solution $\boldsymbol{E}_{k+1}$, $\boldsymbol{F}_{k+1}$ and $\boldsymbol{N}_{k+1}$.

For updating $\boldsymbol{F}$, we utilize Algorithm 1, where $\boldsymbol{f}$, $\boldsymbol{g}$, and $\boldsymbol{z}$ are alternatingly updated within t inner iterations. In our experiment, $t = 10$ is sufficient to guarantee convergence. At each inner iteration, the calculation of $\boldsymbol{f}$ mainly includes one FFT and one inverse FFT, each at a cost of $\mathcal{O}(mnd\log(mnd))$ [38]. The $\boldsymbol{g}$ can be solved by the thresholding operator with a complexity of $\mathcal{O}(mnd)$. To update $\boldsymbol{z}$ only requires a simple algebraic calculation that consumes $\mathcal{O}(mnd)$, which is similar to that of the dual variables $\boldsymbol{O}$ and $\boldsymbol{Q}$. To sum up, the complexity of $\boldsymbol{F}$-subproblem is $\mathcal{O}(tmnd\log(mnd))$.

Finally, the $\boldsymbol{N}$-subproblem can be efficiently solved at a cost of $\mathcal{O}(mnd)$, either in the case of $p = 1$, $1 < p < 2$, or $p = 2$. Therefore, the proposed ℓ_p-LSCE algorithm requires the total complexity of $k\cdot\mathcal{O}(mnd\cdot\min(mn, d) + tmnd\log(mnd))$, where k is the number of outer iteration.

IV. Experimental Results and Discussion

In this section, we evaluate the performance of the proposed ℓ_p-LSCE algorithm with a series of experiments using the moving and stationary target automatic recognition (MSTAR) public database [41]. The database contains various ground military targets. For each class of target, their SAR images are obtained under 15° and 17° depression angles over full 360° aspect angles, respectively. For the ATR experiment, images with 17° depression angle are taken as the training set while those with 15° depression angle are taken as the testing set, which is a typical setting in [1]–[3]. In order to fully evaluate the performance of ATR algorithms, all ten classes of target recognition are carried out in our experiments. These military targets consist of three tank types of T62, T72, and BMP2, two armored car types of BTR60 and BTR70, two cannon types of 2S1 and ZSU23/4, two truck types of BRDM2 and

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TABLE I
TEN CLASSES OF TARGETS IN MSTAR DATABASE FOR THE RECOGNITION

Type	T72	BMP2	BTR70	2S1	BRDM2	BTR60	D7	T62	ZIL131	ZSU23/4
Training (17°)	232 (sn-132)	233 (sn-9563)	233	299	298	256	299	299	299	299
Testing (15°)	196 (sn-132) 195 (sn-812) 191 (sn-s7)	195 (sn-9563) 196 (sn-9566) 196 (sn-c21)	196	274	274	195	274	273	274	274

TABLE II
PARAMETER SETTINGS OF THE PROPOSED ℓ_p-LSCE ALGORITHM IN MSTAR DATABASE

Parameter	λ_1	λ_2	μ_0	η_0	$\mu_{\max}$	ρ	ε
Value	$\frac{3}{\sqrt{mn}}$	$\frac{9}{\sqrt{mn}}$	0.1	0.1	10^7	1.5	0.7

ZIL131, and one bulldozer type of D7. Among target types of T72 and BMP2, there are a total of six subtypes: T72 (sn-132, sn-812, and sn-s7) and BMP2 (sn-9596, sn-9566, and sn-c21). More details about database are summarized in Table I.

The performance of the proposed algorithm is compared with those of the Lee [14], TV [30], SAR-BM3D [19], PPB [20], and joint low-rank and sparse multiview denoising (JLSMD) [11] methods. For all algorithms that participate in the comparison, we reset some parameters instead of using the default settings in original codes to gain the best performance. In detail, we set a proper regularization parameter of $\lambda = 0.1$ for the TV method, and that of $\lambda = 1/\sqrt{mn}$, which is widely suggested in the RPCA model [25], for the JLSMD method. For SAR-BM3D, the only user-defined parameter is the number of looks of the speckle noise, which is reset to 2. Other parameters embedded in the software use default values that are chosen by the authors after rigorous testing. For the PPB method, the α-quantile parameter and filtering parameter are finely tuned to $\alpha = 0.6$ and $T = 0.8$, respectively. For the proposed ℓ_p-LSCE, we flexibly adjust the value of $p \in [1, 2]$ to conduct different regularized schemes, and then test and compare their performance in terms of speckle reduction and target feature enhancement. More details on parameter settings of our proposed algorithm are listed in Table II, where especially two regularization parameters λ_1 and λ_2 are selected by the experiments in Section IV-E.

A. Algorithm Validation

We develop 36 aspect spaces to implement our proposed algorithm by representing each space of 10° aspect angles, respectively. Although the 5° aspect angles are perhaps a better choice, samples of MSTAR database are not dense enough to support a small aspect bin [29]. Fig. 3 shows five original samples from the database as well as their corresponding despeckling results via different algorithms.

We see that two despeckling algorithms of the Lee and SAR-BM3D yield mild results with blurry backgrounds and targets. The TV algorithm produces the sharp texture of target but fails to inhibit strong spots in the background. PPB results in clean and homogeneous structures in the target region. However, the blurry texture and insufficient intensity suppression degrade its despeckling performance in the background region. By exploiting the low-rank information, JLSMD shows an enhanced denoising result but is still influenced by false alarms caused by natural clutters and speckles. Compared with the JLSMD, the proposed ℓ_1-regularized LSCE algorithm further reduces false alarms and yields cleaner target structures, which is attributed to the effective incorporation of low-rank and space-angle continuity information. Nevertheless, the smeared texture in the background shows that it still lacks noise suppression. In order to detect more dense noise component, we apply the ℓ_p-regularized scheme with $p > 1$ for the proposed algorithm. It is observed that using $p = 1.3$ or $p = 1.5$ generates a cleaner background than $p = 1$ while edge features of target are also preserved well.

Apart from the visual comparison, we further quantitatively evaluate the performance of different algorithms by using five assessment criteria, the details of which are listed as follows.

1) *Equivalent Number of Looks (ENL) [23]:* This index evaluates the smooth level of homogeneous region in the SAR image. The general ENL of the image $\boldsymbol{X}$ is defined as follows:

$$\mathrm{ENL} = \frac{\mathrm{E}^2([\boldsymbol{X}]_{i,j})}{\mathrm{Var}([\boldsymbol{X}]_{i,j})} \tag{36}$$

where $\mathrm{E}(\cdot)$ and $\mathrm{Var}(\cdot)$ denote the expectation and variance, respectively. It is worth pointing out that ENL should be measured within the same homogeneous region, where the scene variation can be negligible with respect to the speckle noise fluctuation [22]. Therefore, we preannotate the target region and background region for each image in MSTAR database and then measure the ENL value in the target region (i.e., ENL–T) and the background region (i.e., ENL–B), respectively

$$\begin{aligned}\mathrm{ENL{-}T} &= \frac{\mathrm{E}^2{}_{(i,j)\in\Omega_\mathrm{T}}([\boldsymbol{X}]_{i,j})}{\mathrm{Var}_{(i,j)\in\Omega_\mathrm{T}}([\boldsymbol{X}]_{i,j})} \\ \mathrm{ENL{-}B} &= \frac{\mathrm{E}^2{}_{(i,j)\in\Omega_\mathrm{B}}([\boldsymbol{X}]_{i,j})}{\mathrm{Var}_{(i,j)\in\Omega_\mathrm{B}}([\boldsymbol{X}]_{i,j})}\end{aligned} \tag{37}$$

where Ω_T and Ω_B represent the preannotated target region and background region, respectively.

2) *Target-to-Clutter Ratio (TCR) [42]:* This index measures the relative intensity of target scatterers compared with background clutters, whose definition is given by

$$\mathrm{TCR} = 20\log_{10}\frac{\mathrm{E}_{(i,j)\in\Omega_\mathrm{T}}\left([\boldsymbol{X}]^2_{i,j}\right)}{\mathrm{E}_{(i,j)\in\Omega_\mathrm{B}}\left([\boldsymbol{X}]^2_{i,j}\right)}. \tag{38}$$

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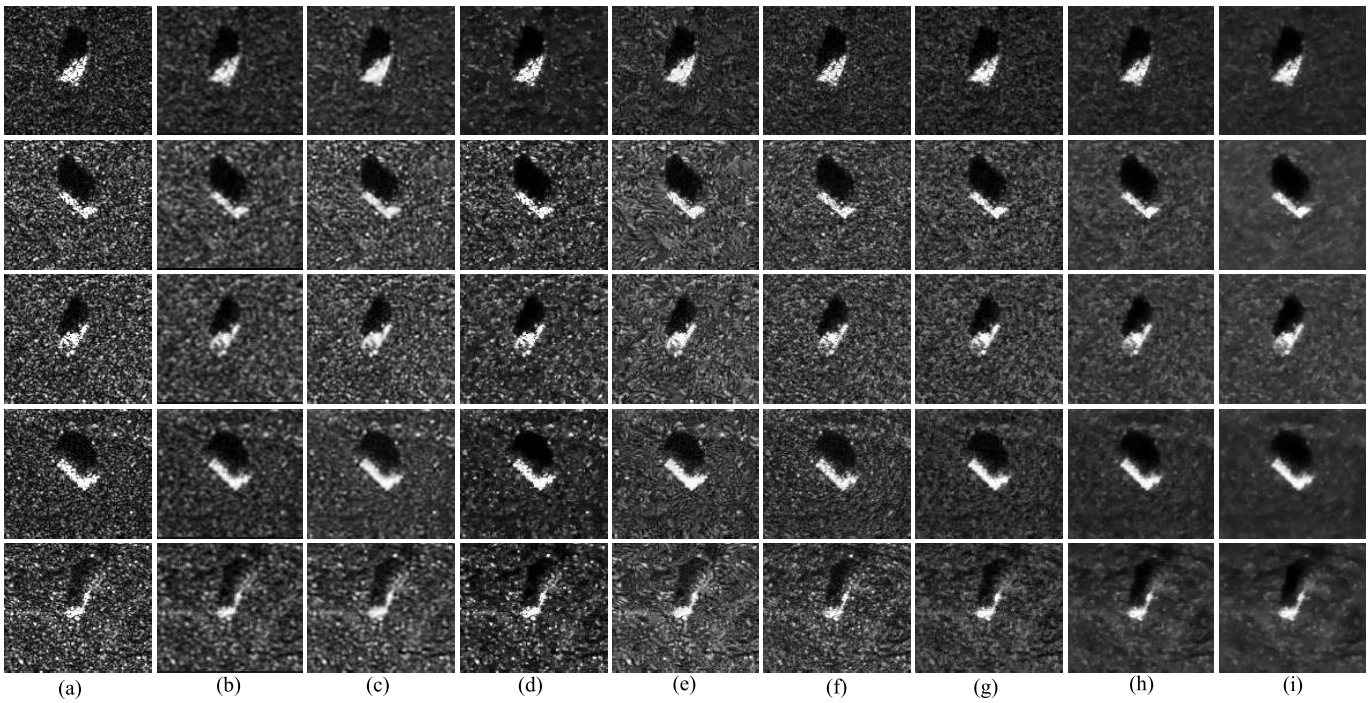


Fig. 3. Comparison results of speckle reduction by different algorithms. Rows 1–5: original images and results from types T72, BTR70, BMP2, 2S1, and BRDM2, respectively.

Note that the larger the TCR is, the better the target enhancement and clutter suppression are.

3) *Mean of Ratio Image (μ_r) [22]:* This index measures the bias between the filtered image and the original image in a point-by-point ratio manner as follows:

$$\mu_r = \mathrm{E}([\boldsymbol{W}]_{i,j}/[\boldsymbol{X}]_{i,j}) \tag{39}$$

where $\boldsymbol{W}$ and $\boldsymbol{X}$ denote the original and filtered images, respectively. It is clear that a measured value closer to one indicates a more unbiased estimation.

4) *Variance of Ratio Image (σ_r^2) [22]:* Similar to μ_r, this index evaluates the bias of estimation and is defined as follows:

$$\sigma_r^2 = \mathrm{Var}([\boldsymbol{W}]_{i,j}/[\boldsymbol{X}]_{i,j}). \tag{40}$$

5) *Image Entropy (IE) [43]:* This index statistically assesses the randomness of image, and the definition is given by

$$\mathrm{IE} = -\sum_{i=0}^{255} x_i \log_2 x_i \tag{41}$$

where x_i represents the appearance probability of the gray level i with a peak value of 255. A smaller entropy value characterizes a cleaner background and target structure with a better noise suppression.

Table III lists the scores of all competing methods in various assessment criteria. Viewing the result of original images as a baseline, we see that all methods yield the larger ENL values. For index ENL–T, Lee and PPB methods show competitive performance while the best performance is achieved by the proposed LSCE algorithm, which verifies its good capability to enhance continuous and homogeneous features in the target region. In addition to ENL–T, the despeckling performance is also evaluated in the background region via ENL–B. With the joint exploitation of low-rank and space-angle continuous properties to separate the speckle noise from SAR images, the proposed LSCE is capable of reducing false alarms effectively and, thus, yields the clean background with remarkable ENL–B values, as shown in Table III. For index TCR, whose values before and after despeckling indicate how much a filter preserves the target radiation characteristic while suppressing the background clutter, the LSCE algorithm obtains the highest score again. For the bias indexes of μ_r and σ_r^2, it is observed that the four methods of SAR-BM3D, PPB, JLSMD, and LSCE yield relatively satisfactory results, along with the best μ_r and σ_r^2 scores achieved by JLSMD and PPB, respectively. Compared with the baseline, we see that all algorithms that participate in the comparison gain the smaller IE values due to their noise suppression. In particular, the proposed LSCE shows a superior robustness against noise to other competing algorithms, as shown in IE values. Finally, comparison of execution times of different methods is reported. We utilize a PC with Intel i7 CPU of 1.8 GHz and RAM of 8 GB, and all methods are conducted in MATLAB. The average execution time for processing each SAR image of size 128×128 is listed in Table III. Note that the JLSMD, which only enforces the low-rank property, provides the fastest processing speed of 0.04 s per image. In contrast, the proposed ℓ_1-LSCE further incorporates the space-angle continuous property to result in an execution time of 0.44 s, which is slightly faster than the ℓ_p-regularized scheme with $1 < p < 2$. The reason lies in that

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TABLE III

QUANTITATIVE COMPARISON OF DIFFERENT SPECKLE REDUCTION METHODS FOR THE FIRST THREE CLASSES OF TARGETS IN MSTAR DATABASE. THE BEST AND SECOND BEST RESULTS UNDER EACH INDEX ARE HIGHLIGHTED IN BOLD AND UNDERLINE, RESPECTIVELY

Index	Type	Original	Compared Methods					Proposed LSCE		
			Lee [14]	SAR-BM3D [19]	TV [30]	PPB [20]	JLSMD [11]	ℓ_1-LSCE	$\ell_{1.3}$-LSCE	$\ell_{1.5}$-LSCE
ENL–T	T72	4.46	8.82	8.52	4.91	<u>9.17</u>	7.14	7.89	8.96	**9.46**
	BMP2	5.30	11.84	11.19	5.87	<u>12.39</u>	8.92	9.67	11.17	**12.72**
	BTR70	4.21	<u>8.91</u>	8.69	4.72	8.84	6.81	7.41	8.29	**9.11**
ENL–B	T72	2.38	6.04	3.30	3.13	5.16	5.27	6.55	<u>15.86</u>	**20.75**
	BMP2	2.39	6.25	3.48	2.79	5.18	5.42	7.52	<u>13.90</u>	**20.27**
	BTR70	2.49	6.77	3.76	3.03	5.60	5.70	8.04	<u>14.62</u>	**21.46**
TCR	T72	22.18	23.41	22.01	24.94	22.74	22.89	25.05	**25.66**	<u>25.35</u>
	BMP2	17.83	19.06	17.69	19.82	18.25	18.58	20.87	**21.53**	<u>21.04</u>
	BTR70	16.27	17.08	16.50	17.93	17.37	16.93	18.30	**19.32**	<u>18.98</u>
μ_r	T72	1	1.07	<u>0.95</u>	1.24	0.90	**1.04**	1.07	1.08	1.08
	BMP2	1	1.06	<u>0.95</u>	1.25	0.89	**1.03**	1.06	1.08	1.07
	BTR70	1	1.06	0.94	1.22	0.89	**1.03**	<u>1.05</u>	1.07	1.06
σ_r^2	T72	0	1.82	<u>0.55</u>	2.43	**0.51**	0.72	0.60	0.65	0.75
	BMP2	0	1.93	0.57	2.23	**0.48**	0.77	<u>0.52</u>	0.72	0.67
	BTR70	0	1.59	<u>0.51</u>	2.18	**0.48**	0.72	0.55	0.71	0.64
IE	T72	6.77	6.31	6.22	6.06	6.18	6.36	6.03	<u>5.50</u>	**5.30**
	BMP2	7.14	6.66	6.75	6.12	6.63	6.65	6.04	<u>5.97</u>	**5.76**
	BTR70	7.20	6.72	6.65	6.14	6.79	6.74	6.17	<u>6.03</u>	**5.83**
Time (s)	/	/	0.51	4.23	<u>0.05</u>	0.55	**0.04**	0.44	0.74	0.71

the ℓ_p-regularized scheme can obtain the closed-form solution to the $\boldsymbol{N}$-subproblem when $p = 1$, but not when $1 < p < 2$ since an iterative procedure for root finding is required in this case. Overall, the proposed LSCE gains the best scores in four of the seven assessment criteria, as listed in Table III.

B. ATR Validation

With the increasing demand of ATR problem, an enhanced feature of sample plays a significant role in improving the recognition performance. Although using the primary sample as the feature is a common way in many ATR algorithms, the presence of speckle and background clutter increases the probability of false alarm and, hence, degrades the ATR performance. To improve the recognition accuracy, the proposed algorithm and other comparison algorithms are applied in the image preprocessing stage. Then, we use the resulting images as enhanced features to be fed into the subsequent ATR algorithms. Since the Lee, SAR-BM3D, TV, and PPB algorithms are conducted on the single image, we use them to enhance each training sample and testing sample, respectively. To ensure a fair comparison, we also implement the JLSMD and our proposed algorithm on both training set and testing set. For the training set, the feature-enhanced speckle reduction is applied to training images class by class. Since the class labels of testing samples are not known *a priori*, the classwise speckle reduction for the testing set is not feasible, and thus, we separate it from the speckle reduction in the training set as follows. We implement two algorithms by mixing all classes of testing samples together in the same aspect space. To be clearer, for the developed 36 aspect spaces in the testing set, each space contains samples of the corresponding aspect angles across all classes. Note that no test images were used for training; thus, our implementation conforms to the specification of the target recognition task.

Here, we test several representative ATR algorithms, including the support vector machine (SVM) [1], adaptive boosting (AdaBoost) [2], collaborative representation-based k-nearest neighbor (CR-based k-NN) [4], and sparse representation classification (SRC) [5] algorithms. Dictionary learning-based methods, such as the K-singular value decomposition (K-SVD) [9] and label constraint K-SVD (LC-KSVD) [10], are also involved. To ensure fairness, all dictionary learning methods have the sparsity factor of 20. The number of dictionary atoms is set to 600.

With respect to each ATR algorithm, we test different feature enhancement methods whose achieved recognition accuracies are listed in Table IV. It is clear that all methods improve the recognition performance, compared with the baseline result by using original images. Especially, the proposed ℓ_p-LSCE algorithm shows remarkable superiority to other comparison methods under various classifiers. Due to the effective feature enhancement for the original images, LSCE improves the accuracy of CR-based k-NN classifier by nearly 8% and SVM classifier by nearly 5%. Under the SRC classifier, the proposed LSCE algorithm achieves the highest accuracy of up to 98.92% when using the ℓ_p-regularized scheme with $p = 1.4$. It is worth mentioning that in recent years, emerging deep learning-based methods have been proved to be successful for SAR ATR problems. Here, we also compare the recognition performance of our method with that of several state-of-the-art deep learning algorithms for the same ten-class ATR problem

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TABLE IV

RECOGNITION ACCURACIES ACHIEVED USING DIFFERENT FEATURE ENHANCEMENT METHODS, COMPRISING LEE [14], SAR-BM3D [19], TV [30], PPB [20], JLSMD [11], AND THE PROPOSED ℓ_p-LSCE. THE RESULTS OBTAINED USING ORIGINAL SAMPLES CAN BE VIEWED AS BASELINES. THE BEST AND SECOND BEST RESULTS UNDER EACH CLASSIFIER ARE HIGHLIGHTED IN BOLD AND UNDERLINE, RESPECTIVELY

Method	Recognition accuracy (%)									
	Original	Lee	SAR-BM3D	TV	PPB	JLSMD	ℓ_1-LSCE	$\ell_{1.3}$-LSCE	$\ell_{1.7}$-LSCE	ℓ_2-LSCE
SVM [1]	93.23	95.38	94.66	95.57	95.42	96.23	96.62	<u>97.89</u>	**98.03**	97.31
AdaBoost [2]	92.84	94.47	94.05	94.84	95.23	95.87	96.91	**97.56**	<u>97.33</u>	96.62
CR-based k-NN [4]	90.26	92.85	92.91	92.60	92.88	93.19	95.79	97.33	**97.86**	<u>97.56</u>
SRC [5]	95.85	97.82	97.78	97.25	97.70	98.15	98.44	**98.89**	<u>98.75</u>	97.91
K-SVD [9]	94.41	95.10	95.64	95.21	96.13	97.24	<u>97.35</u>	**97.82**	97.00	95.79
LC-KSVD1 [10]	95.32	95.53	95.38	95.10	96.01	97.72	<u>97.90</u>	**98.32**	97.45	96.84
LC-KSVD2 [10]	94.90	95.47	95.62	94.91	95.64	97.42	97.48	**98.05**	<u>97.63</u>	96.43

on the MSTAR database. In [6], a multiview deep learning framework is designed to combine SAR images with aspect angles to acquire a recognition accuracy of 98.52%. In [3], the all-convolutional networks (A-ConvNets) is proposed by using the sparsely connected convolution architectures. With the data augmentation to expand the training data set, A-ConvNet yields the recognition rate of 99.13%, slightly higher than our method. In [7], the methods of deep learning and sparse representation are incorporated to achieve a fusion-based recognition result of 99.83%, which is 0.94% higher than our method.

In addition to using primary images, we also test the ATR performance in the presence of speckle noise. We corrupt all training and testing images by the noise at SNR $=$ 15 dB (see Section IV-D for details on noise addition), and then apply our method and other competing methods to the image preprocessing stage of ATR experiment. Table V lists the recognition accuracies in the noisy environment. It is noted that our method acquires the highest accuracy of up to 97.16% under the SRC classifier when using the ℓ_p-regularization of $p = 1.7$ among four different regularized schemes. Moreover, our method significantly improves the accuracy of the CR-based k-NN classifier by more than 10%, compared with the result by using original images.

C. Effects of ℓ_p-Regularized Schemes

We investigate how the selection of ℓ_p-regularized scheme affects the proposed LSCE algorithm. Note that the optimal choice of $p \in [1, 2]$ relies on the statistical characteristics of noise and, hence, is case-dependent. In the presence of Gaussian noise, the ℓ_2-regularization is statistically optimal and has good capability to smooth the homogeneous region [44]. Instead, if the noise is with a spike-like property, which is somewhat analogous to sparsity, the choice of $p = 1$ outperforms $p = 2$ due to its less sensitivity to outliers [44]. Generally speaking, a smaller p is suitable for a more sparse noise distribution. For MSTAR database, speckles and natural clutters are widespread in SAR images, where the noise shows a moderately sparse distribution and, thus, the choice of $1 < p < 2$ is statistically appropriate. In addition to noise suppression, we also need to determine a proper regularized scheme from the perspective of target structure preservation. As the value of p increases from 1 to 2, the homogeneous region could be better smoothed due to a denser noise capture; however, it is accompanied by the inevitable degradation of target details, such as edges and sharp textures. To summarize, the selection of regularized scheme is a tradeoff between denoising and preserving target details.

Fig. 4 shows the comparison of different ℓ_p-regularized schemes by their extracted target components as well as noise components from two original images of types D7 and BRDM2. Note that JLSMD also performs speckle reduction with the noise component separated from the target component, and thus, we include it in the comparison. Compared with JLSMD that only detects strong speckles, it is clearly observed that LSCE achieves the more sufficient noise suppression. Especially, the ℓ_p-regularized scheme with a larger value of p is validated to have a superior robustness against noise. As shown in Fig. 4, the regularized scheme with $1 < p < 2$ can extract a more dense and salient noise component than that with $p = 1$ and, hence, achieves the better speckle reduction for the target component. Although the case of $p = 2$ yields a higher contrast between the background and target, it also degrades the resolution of target and shows oversmoothed edges and textures in resulting images.

Then, we conduct the experiment to investigate the effect of ℓ_p-regularized scheme on ATR performance. Fig. 5(a) illustrates the recognition accuracy versus p under different classifiers using original images, while Fig. 5(b) exhibits the results using corrupted images by the speckle noise at SNR $=$ 15 dB. The curves exhibit that the highest recognition rate of each classifier corresponds to a certain value of $1 < p < 2$. In contrast to $p = 1$, we can conclude that using a proper ℓ_p-regularized scheme with $1 < p < 2$ can achieve the better ATR performance. Of particular note is that with regard to the same classifier, the best choice of p when using original images for ATR might be different from that when using corrupted images for ATR. Taking the SRC classifier as an example, the optimal p that results in the highest recognition rate increases from $p = 1.4$ in Fig. 5(a) to $p = 1.7$ in Fig. 5(b). We see that the increase of optimal value of p also occurs under SVM, K-SVD, and CR-based k-NN classifiers. The reason lies in that the optimal regularized

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TABLE V

RECOGNITION ACCURACIES ACHIEVED USING DIFFERENT FEATURE ENHANCEMENT METHODS, COMPRISING LEE [14], SAR-BM3D [19], TV [30], PPB [20], JLSMD [11], AND THE PROPOSED ℓ_p-LSCE, IN THE SPECKLE NOISE. THE RESULTS OBTAINED USING ORIGINAL SAMPLES CAN BE VIEWED AS BASELINES. THE BEST AND SECOND BEST RESULTS UNDER EACH CLASSIFIER ARE HIGHLIGHTED IN BOLD AND UNDERLINE, RESPECTIVELY

Method	Recognition accuracy (%)									
	Original	Lee	SAR-BM3D	TV	PPB	JLSMD	ℓ_1-LSCE	$\ell_{1.3}$-LSCE	$\ell_{1.7}$-LSCE	ℓ_2-LSCE
SVM [1]	90.04	94.66	91.07	91.82	93.47	91.32	93.01	<u>95.60</u>	**95.62**	95.31
AdaBoost [2]	89.51	93.67	92.65	91.45	94.19	92.37	93.62	**95.46**	<u>95.08</u>	94.86
CR-based k-NN [4]	85.45	90.54	90.24	86.79	91.08	91.23	92.49	94.10	<u>96.29</u>	**96.35**
SRC [5]	92.26	96.32	95.64	93.12	95.85	93.41	95.51	<u>96.88</u>	**97.16**	96.80
K-SVD [9]	88.46	94.15	93.17	91.07	93.94	92.20	93.32	**94.84**	<u>94.13</u>	93.21
LC-KSVD1 [10]	88.29	93.84	93.88	90.46	<u>94.75</u>	93.14	94.23	**95.26**	94.56	93.47
LC-KSVD2 [10]	87.67	93.79	93.83	90.05	93.76	92.82	93.33	**94.67**	<u>94.23</u>	93.14

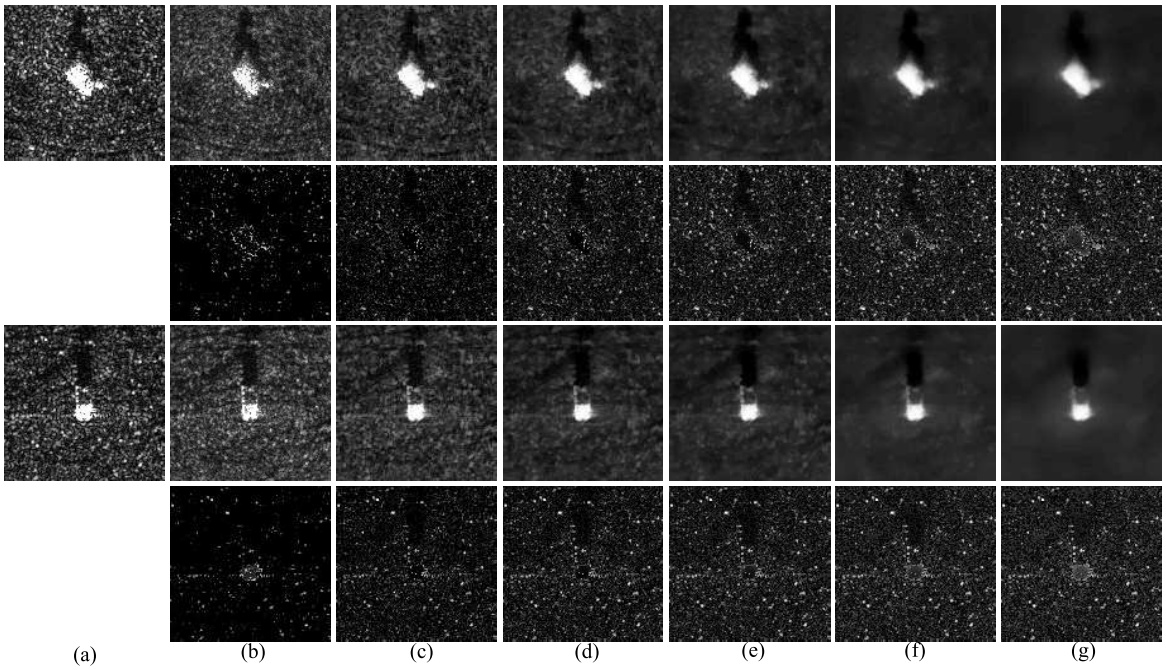


Fig. 4. Comparison results of speckle reduction by different algorithms. The first and third rows: original images as well as despeckled target components from original images. The second and fourth rows: detected noise components from original images.

scheme depends on the noise distribution. In corrupted scenarios, where the interference is more severe than that in original scenarios, the proposed algorithm requires a greater degree of regularization with a larger p to filter out the speckle noise. Therefore, it can be concluded that a flexible and appropriate adjustment of p helps to handle varied scenarios to improve the ATR performance.

D. Robustness Against Noise

Next, the noise robustness of the proposed algorithm is investigated. We corrupt original SAR images by the speckle noise, which has a multiplicative property and can be viewed as the product of RCS of object and stochastic noise within the synthetic aperture time [13], [14]. As suggested in [32], we use the Gamma distribution with the shape parameter of $\alpha = 2$ to model the speckle noise, and then add the noise at SNR $= 15$ dB into all training and testing images for the ATR experiment. Fig. 6 shows three examples of the original and noisy images. It is worth pointing out that the third example illustrates a challenging scenario, where the target texture is severely damaged by widespread false alarms.

The denoising results of speckle noise via different algorithms are also displayed in Fig. 6. We see that the LSCE algorithms have strong robustness against noise, even under the challenging scenario. Compared with the single-image denoising algorithms, the LSCE jointly exploits the low-rank and continuity knowledge between images, which is demonstrated to restore the target structure. Although the JLSMD also incorporates the low-rank structure in adjacent images, its results are still contaminated by strong noise spots and exhibit indistinct boundaries between the target and background. Compared with JLSMD, the proposed LSCE further enforces the space-angle continuity with a flexible

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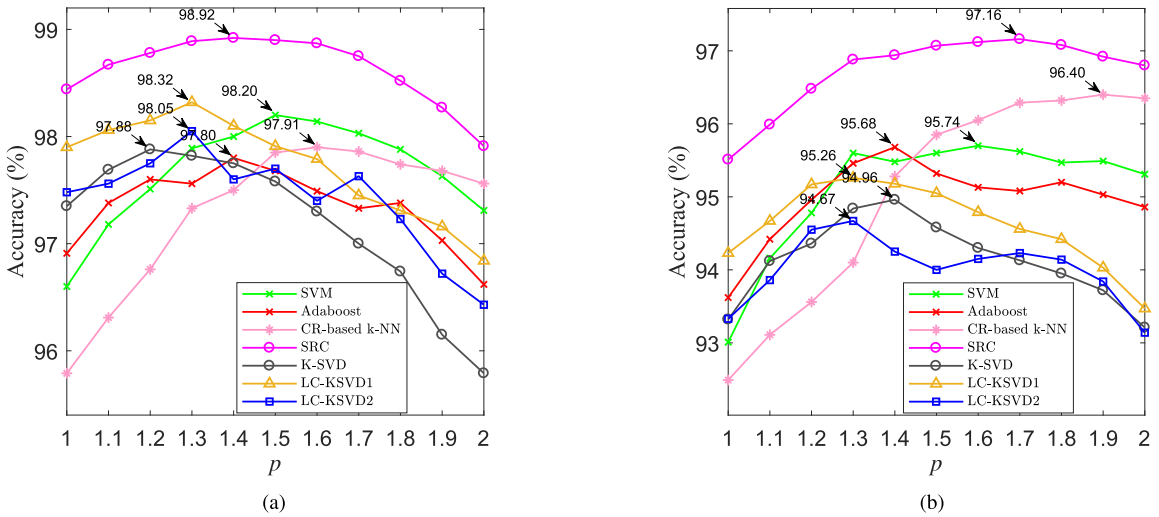


Fig. 5. Recognition accuracy versus p when using the proposed ℓ_p-LSCE, with different ℓ_p-regularized schemes, to enhance (a) original images and (b) corrupted images by the speckle noise at SNR = 15 dB. The arrows in the picture mark the ℓ_p-regularized scheme with the highest recognition rate under each classifier.

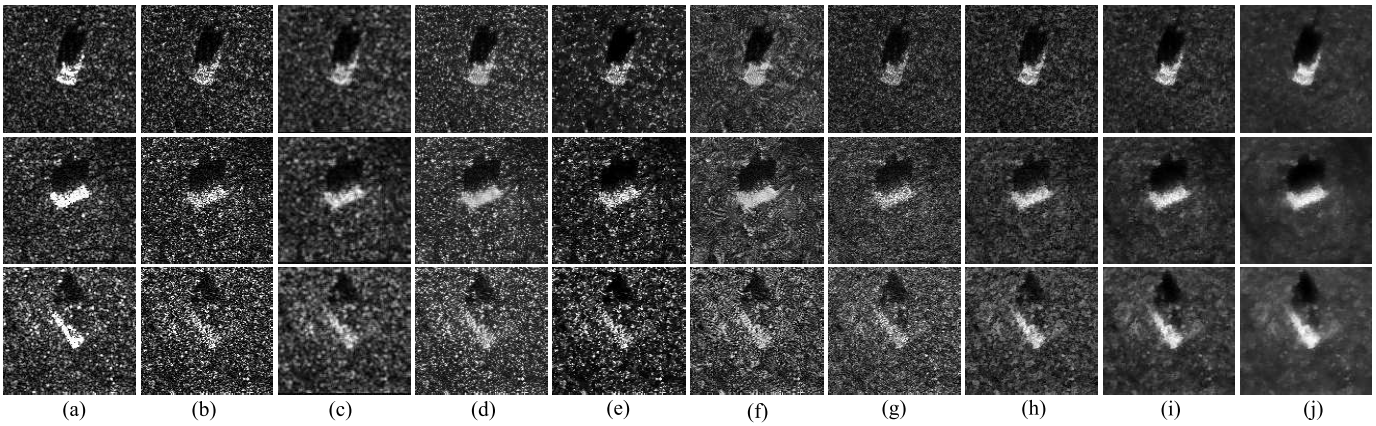


Fig. 6. Comparison results of speckle noise reduction by different algorithms. Rows 1–3: results of images from types T72, T62, and ZIL131, respectively.

ℓ_p-regularized scheme to handle various scenarios, whose remarkable noise resistance can be found in Fig. 6. Numerical experiments clearly demonstrate that the joint enforcement of ℓ_p-regularized low-rank and space-angle continuity constraint can not only enhance the target but also filter out the noise.

E. Sensitivity Analysis and Discussion of Limitations

1) Sensitivity to the Regularization Parameters: In the proposed algorithm, there are two regularization parameters of λ_1 and λ_2 to trade off the low-rank constraint and space-angle continuity constraint. A similar consideration of regularization parameter occurs in the well-known RPCA model in (2), where the typical setting $\lambda = 1/\sqrt{mn}$ is recommended to guarantee the existence of optimal solution [25]. Note that our model is different from RPCA due to the additional space-angle continuity constraint; thus, we set $\lambda_1 = c_1/\sqrt{mn}$ and $\lambda_2 = c_2/\sqrt{mn}$ with the tuning parameters c_1 and c_2 for the proposed ℓ_p-LSCE algorithm. Fig. 7 displays the variations of ATR performance when jointly taking these two tuning parameters into account, where c_1 and c_2 vary in the set $\{2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6\}$ and $\{3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$, respectively. Here, the SRC classifier is used to perform the ATR after feature enhancement due to its highest recognition accuracy among all classifiers reported in Section IV-B. As illustrated in Fig. 7(a) and (b), the ATR performance varies gradually with the change of two regularization parameters, whose appropriate choices contribute to the balance between different constraint terms in our model to improve the recognition rate. We find that the best performance is achieved at $(c_1, c_2) = (3, 9)$. Therefore, the uses of $\lambda_1 = 3/\sqrt{mn}$ and $\lambda_2 = 9/\sqrt{mn}$ are suggested in the experiments.

2) Sensitivity to the Aspect Angle: Because the proposed LSCE performs the multiview speckle reduction using images from the same pose space of aspect angles, we then investigate how the selection of aspect angle increment used for dividing different spaces affects the performance. Due to the scattering mechanism, SAR images captured within 5° aspect angles are usually stable and, thus, highly interrelated [29]. However, if the angular increment is too small, MSTAR database will not be dense enough for a pose space with multiple images, thus degrading the interrelated information used for the multiview

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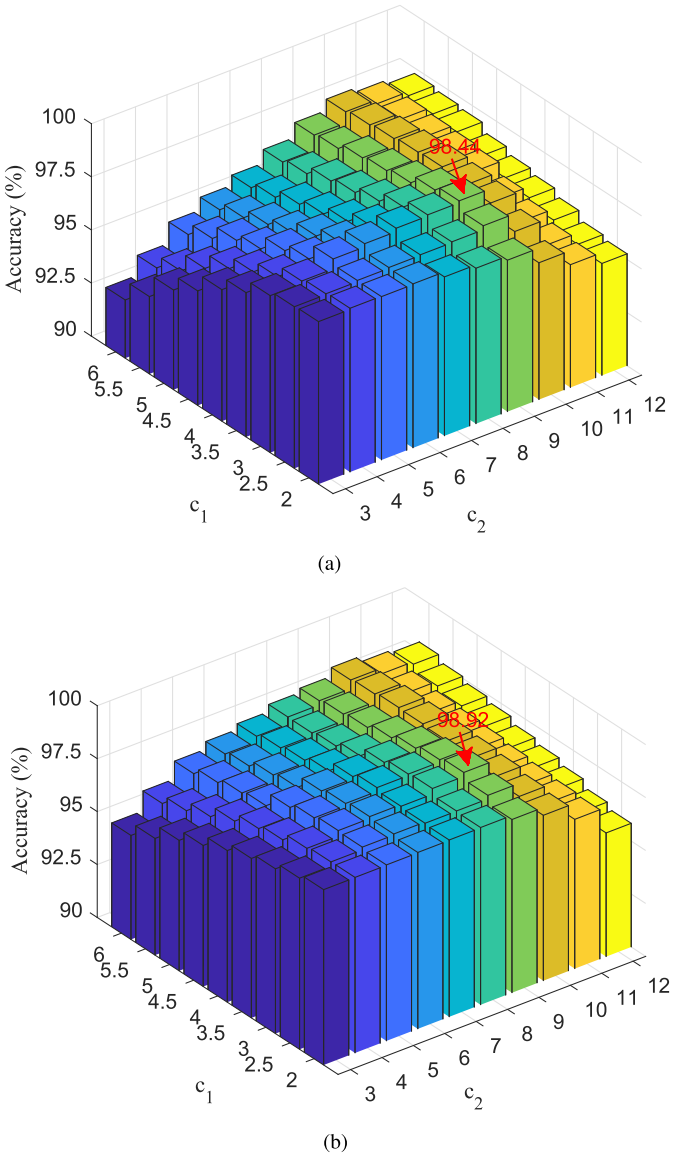


Fig. 7. Recognition accuracy versus two regularization parameters under SRC classifier when using the proposed LSCE with (a) ℓ_1-regularized scheme and (b) $\ell_{1.4}$-regularized scheme.

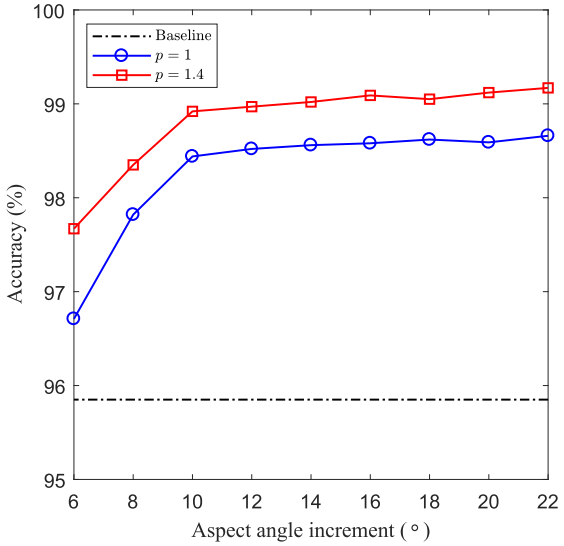


Fig. 8. Recognition accuracy versus aspect angle increment under SRC classifier when using the proposed ℓ_p-LSCE with $p = 1$ and $p = 1.4$. Note that the baseline plots the result of using original images for the ATR.

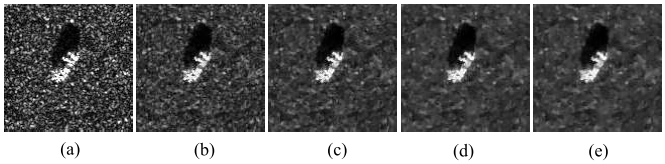


Fig. 9. Visual comparison of speckle reduction by using different aspect angle increments for dividing pose spaces. (a) Original. (b) 6°. (c) 8°. (d) 10°. (e) 16°.

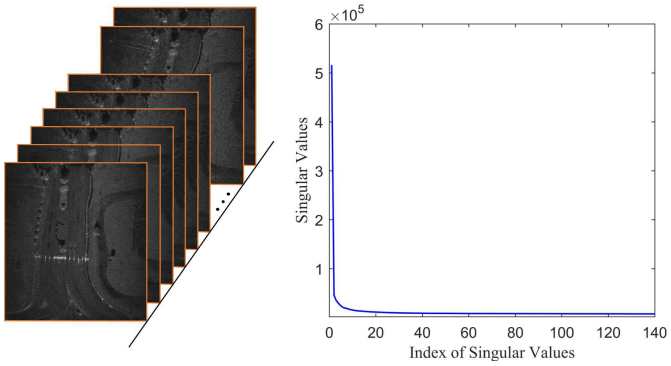


Fig. 10. Illustration of the Eubank Gate and Traffic VideoSAR sequence. (Right) Singular values of the first 200 frames of video sequence.

despeckling. Fig. 8 shows the recognition accuracy of ℓ_p-LSCE versus the aspect angle increment under SRC classifier, where a baseline result of using original images for the ATR is also plotted for the guide of comparison. We can clearly see that the LSCE algorithm is somewhat robust to the selection of aspect angle increment when its value is greater than 10°. Nevertheless, in the case of 8° or even 6° angular increment, the ATR performance degrades along with the decrease of number of images in a pose space. More visual results about the effect of aspect angle increment are shown in Fig. 9. Compared with the case of small increment of 6°, we see that a choice of 10° or 16° increment provides a better denoising result and, hence, achieves the recognition improvement.

3) Discussion of Limitations: Although the proposed LSCE could enhance the despeckling performance via low-rank and space-angle continuity, the improvement is at the expense of computational burden. As reported in Section IV-A, the execution time of LSCE is inferior to that of JLSMD, which only uses the low-rank property to perform the RPCA model and, thus, leads to a more tractable optimization problem. Because our image formation model is based on the 3-D tensor structure, the enforcement of 3-D TV constraint for encoding the space-angle continuity inevitably increases the dimension in resultant optimization problem. As listed in Table III, our method is more computationally intensive compared with the conventional 2-D TV denoising. In addition to computational complexity, the proposed LSCE requires the *a priori* knowledge of aspect

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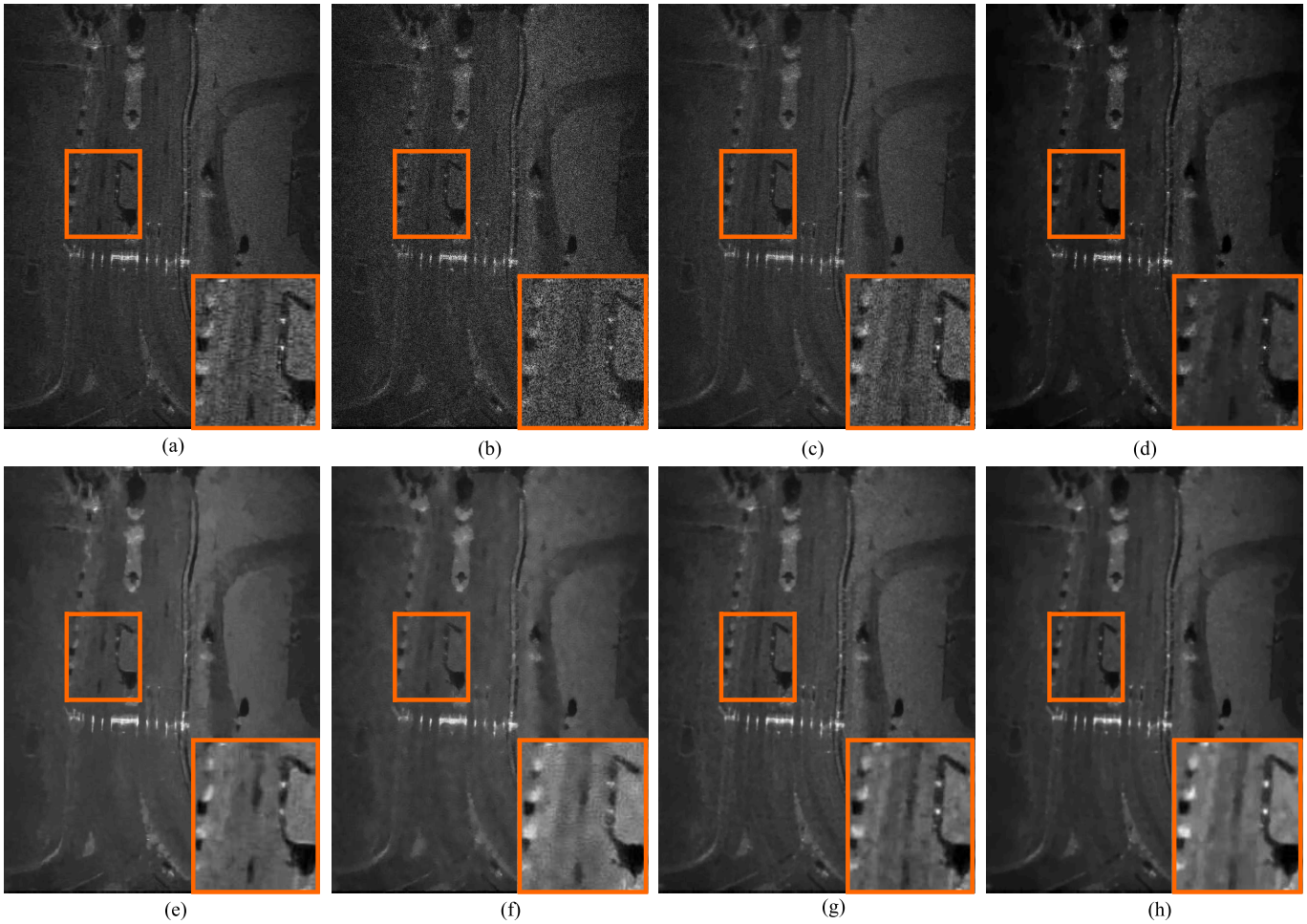


Fig. 11. Visual comparison of speckle reduction in the first image frame from the Eubank Gate and Traffic VideoSAR sequence. (a) Original image is contaminated by (b) speckle noise at SNR = 10 dB, then it is despeckled by (c) JLSMD [11], (d) TV [30], (e) SAR-BM3D [19], (f) PPB [20], (g) proposed ℓ_1-LSCE, and (h) proposed $\ell_{1,3}$-LSCE.

angle for the multiview denoising. From the perspective of ATR problem, it implies that the improvement of recognition accuracy is attributed to the extra angular information of samples. With respect to a new testing sample to be identified, we must know which pose space its aspect angle belongs to so as to carry out the feature enhancement, which might be unavailable under certain scenarios.

In addition to MSTAR database, we also utilize the proposed algorithm to perform speckle reduction on VideoSAR database[1], which includes the high-resolution footage of the Eubank Gate and Traffic at the Kirtland Air Force Base. In VideoSAR mode, the motion of aircraft platform causes continuous variations in the observing angle, thereby continuously collecting SAR image frames with interframe geometric rotation. The geometric rotation between frames offers the highly similar contents between adjacent images. Hence, the low-rank structure resided in the video sequence is notable. As shown in Fig. 10, we arrange different frames of video in order and then stack them column by column into a matrix, whose singular values are also plotted in Fig. 10 and exhibit a heavy-tailed distribution. Note that the number of singular values larger than 5% of the maximum is 5, which accounts for only a small fraction of all singular values, and hence, the matrix has a natural low-rank property.

Here, we implement the speckle reduction using the first 20 frames of video, whose original versions are further corrupted by the speckle noise at SNR = 10 dB. Then, different methods are applied for speckle reduction from noisy images and the quantitative comparison is listed in Table VI. Depeckling results of the first frame are exhibited in Fig. 11, where the result of the Lee method is not included due to its poor visual performance. TV method produces sharp edges and textures in the resulting image but cannot enhance homogeneous regions sufficiently. JLSMD uses the interrelationship between frames to perform the multiview speckle reduction. By exploiting the low-rank information in video sequence, it yields satisfactory denoising results but is inferior to the SAR-BM3D and PPB methods. Our method takes the advantage of the low-rank and continuous knowledge between frames to resist the random noise effect in each frame. Compared with SAR-BM3D and PPB, the proposed LSCE method shows the superior image detail preservation. As shown in Fig. 11, our method recovers

[1]https://www.sandia.gov/radar/video/

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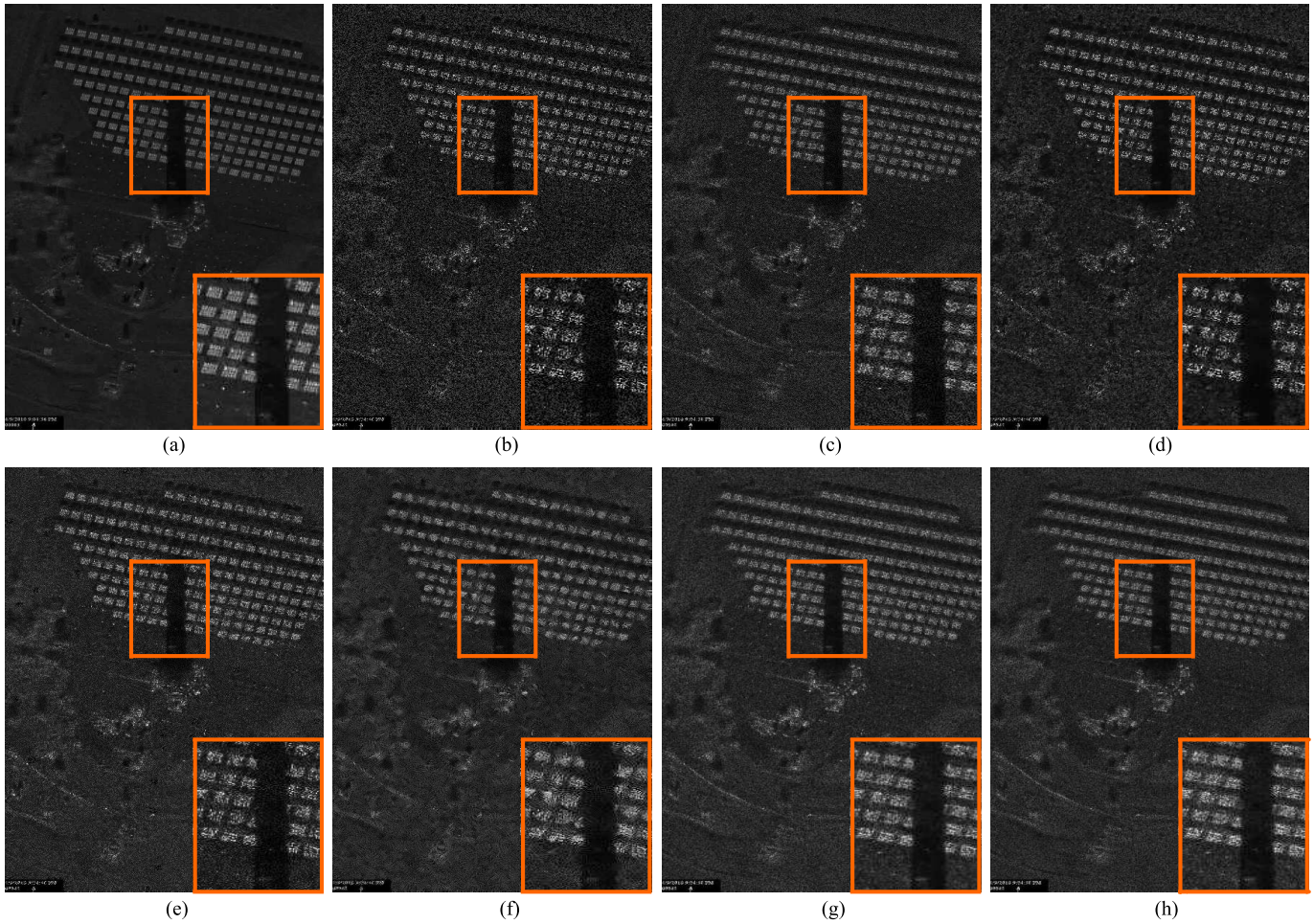


Fig. 12. Visual comparison of speckle reduction in the first image frame from the Solar Tower VideoSAR sequence. (a) Original image is contaminated by (b) speckle noise at SNR = 0 dB, then it is despeckled by (c) JLSMD [11], (d) TV [30], (e) SAR-BM3D [19], (f) PPB [20], (g) proposed ℓ_1-LSCE, and (h) proposed $\ell_{1.3}$-LSCE.

TABLE VI

QUANTITATIVE COMPARISON OF DIFFERENT SPECKLE REDUCTION METHODS FOR THE FIRST 20 FRAMES IN THE EUBANK GATE AND TRAFFIC VIDEOSAR SEQUENCE. THE BEST RESULT UNDER EACH INDEX IS HIGHLIGHTED IN BOLD

Method	ENL	μ_r	σ_r^2	IE
Lee [14]	9.09	1.07	0.40	6.00
JLSMD [11]	11.37	**1.05**	0.32	5.79
TV [30]	7.03	1.24	0.45	5.89
PPB [20]	14.41	0.85	0.15	6.06
SAR-BM3D [19]	14.52	0.82	0.08	6.09
ℓ_1-LSCE	14.79	0.82	**0.04**	6.07
$\ell_{1.3}$-LSCE	**14.96**	0.84	0.05	**5.14**

edges of Eubank gate and structures of strong scatterers well while smoothing homogeneous regions precisely.

F. Extension to VideoSAR Database for Speckle Reduction

Next, we utilize another high-resolution VideoSAR footage of the Solar Tower at Sandia to further compare the despeckling performance of different algorithms. Using the original video frames as the ground truths, we contaminate them by the speckle noise at SNR = 0 dB. Fig. 12 shows the speckle reduction results of the first frame, and we see that the proposed LSCE produces fine details about the tower and surrounding objects. For the quantitative evaluation, four without-reference metrics of ENL, μ_r, σ_r^2, and IE are measured via the average results of all frames. Apart from without-reference metrics, three with-reference metrics are also adopted for evaluation as follows.

1) *Peak Signal-to-Noise Ratio (PSNR) [22]:* This index measures the bias between the despeckled image $\boldsymbol{X}$ and the ground truth $\boldsymbol{G}$ as follows:

$$\mathrm{PSNR} = 10\log_{10}\left(f_{\mathrm{peak}}^2/\mathrm{E}(([\boldsymbol{X}]_{i,j} - [\boldsymbol{G}]_{i,j})^2)\right) \quad (42)$$

where f_{peak} represents the maximum value allowed by the samples dynamic range.

2) *Mean Structural Similarity Index Measurement (MSSIM) [45]:* This index assesses the structural change before and after denoising as follows:

$$\mathrm{MSSIM} = \frac{1}{M}\sum_{k=1}^{M}\mathrm{SSIM}(\mathbf{x}_k, \mathbf{g}_k) \quad (43)$$

where M denotes the number of local windows of image, $\mathbf{x}_k$ and $\mathbf{g}_k$ represent the image patch of $\boldsymbol{X}$ and $\boldsymbol{G}$ at

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TABLE VII
QUANTITATIVE COMPARISON OF DIFFERENT SPECKLE REDUCTION METHODS FOR THE FIRST 20 FRAMES IN THE SOLAR TOWER VIDEOSAR SEQUENCE. THE BEST RESULT UNDER EACH INDEX IS HIGHLIGHTED IN BOLD

Method	ENL	μ_r	σ_r^2	IE	PSNR	MSSIM	EC
Lee	4.29	1.19	3.81	6.24	21.13	0.52	0.03
JLSMD	4.70	1.18	4.10	6.13	22.21	0.63	0.08
TV	2.07	1.20	5.21	6.08	22.05	0.59	0.19
PPB	3.31	**0.87**	4.27	6.46	21.42	0.58	0.05
SAR-BM3D	4.83	0.79	0.67	5.94	22.36	**0.72**	0.21
ℓ_1-LSCE	5.20	0.80	**0.53**	6.00	22.84	0.68	0.21
$\ell_{1.3}$-LSCE	**5.98**	0.78	0.57	**5.80**	**22.99**	0.70	**0.23**

the kth local window, respectively, and the structural similarity index measurement is given by

$$\mathrm{SSIM}(\mathbf{x},\mathbf{g}) = \frac{\big(2\mathrm{E}(\mathbf{x})\mathrm{E}(\mathbf{g})+C_1\big)\big(2\,\mathrm{Cov}(\mathbf{x},\mathbf{g})+C_2\big)}{\big(\mathrm{E}(\mathbf{x}^2)+\mathrm{E}(\mathbf{g}^2)+C_1\big)\big(\mathrm{Var}(\mathbf{x})+\mathrm{Var}(\mathbf{g})+C_2\big)} \tag{44}$$

where $C_1 = K_1^2 f_{\text{peak}}^2$ and $C_2 = K_2^2 f_{\text{peak}}^2$ are two constants with $K_1 \ll 1$ and $K_2 \ll 1$, respectively.

3) *Edge Correlation (EC) [46]:* This index evaluates the edge preservation quality in denoising and is defined as follows:

$$\mathrm{EC} = \frac{\mathrm{Cov}([\boldsymbol{X}_{\mathrm{h}}]_{i,j},[\boldsymbol{G}_{\mathrm{h}}]_{i,j})}{\sqrt{\mathrm{Var}\left([\boldsymbol{X}_{\mathrm{h}}]_{i,j}\right)\cdot\mathrm{Var}\left([\boldsymbol{G}_{\mathrm{h}}]_{i,j}\right)}} \tag{45}$$

where $\mathrm{Cov}(\cdot)$ denotes the covariance, and $\boldsymbol{X}_{\mathrm{h}}$ and $\boldsymbol{G}_{\mathrm{h}}$ are the highpass-filtered versions of $\boldsymbol{X}$ and $\boldsymbol{G}$, respectively, obtained with a 3×3 pixel standard approximation of Laplacian operator.

Quantitative comparison of different algorithms is listed in Table VII. It is obviously clear that the proposed LSCE yields the highest scores in five of the seven metrics, which demonstrates its efficiency and robustness to speckle noise.

V. CONCLUSION

In this article, a feature-oriented speckle reduction algorithm is proposed to enhance the target structure while suppressing the interference for the SAR image. Combining the low-rank property and space-angle continuity, the proposed ℓ_p-LSCE algorithm is capable to enhance target features with the separation of speckle noise, which is conducive to improving the subsequent recognition task. Furthermore, the unified ℓ_p-regularized framework is devised to enhance the noise robustness. The major advantage of our proposed algorithm is its capability to use the interrelation between images for speckle reduction, which is different from many existing methods based on the single-image despeckling. However, it also implies that the proposed algorithm only works for the circular SAR since multiple interrelated images are required for processing. Another inspiring heuristics is that our image formation model extends the conventional 2-D target structure to the 3-D tensor structure, which promotes the investigation of the target structure constraint on the unified space-angle dimensions. The experimental results of circular SAR databases of MSTAR and VideoSAR demonstrate the superiority of the proposed ℓ_p-LSCE algorithm over several representative methods in terms of speckle reduction, target feature enhancement, and ATR accuracy improvement. Our future research is concentrated toward the target structure representation under the 3-D image formation model. Besides the proposed low-rank property and space-angle continuity, one can take account of other priori knowledge to represent the underlying target features. Another open question is whether one can accelerate the 3-D despeckling processing from sampled measurements, which is viewed as a powerful strategy in high-dimensional data optimization [47]. If the proposed 3-D image formation model can be extended to the one that only requires the sampled data to perform speckle reduction, the computational efficiency of the algorithm will be greatly improved.

REFERENCES

[1] Q. Zhao and J. C. Principe, "Support vector machines for SAR automatic target recognition," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 2, pp. 643–654, Apr. 2001.

[2] Y. Sun, Z. Liu, S. Todorovic, and J. Li, "Adaptive boosting for SAR automatic target recognition," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 43, no. 1, pp. 112–125, Jan. 2007.

[3] S. Chen, H. Wang, F. Xu, and Y.-Q. Jin, "Target classification using the deep convolutional networks for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4806–4817, Aug. 2016.

[4] W. Li, Q. Du, F. Zhang, and W. Hu, "Collaborative-Representation-Based nearest neighbor classifier for hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 2, pp. 389–393, Feb. 2015.

[5] J. Wright, A. Yang, A. Ganesh, S. Sastry, and Y. Ma, "Robust face recognition via sparse representation," *IEEE Trans. Pattern Mach. Intell.*, vol. 31, no. 2, pp. 210–227, Jun. 2009.

[6] J. Pei, Y. Huang, W. Huo, Y. Zhang, J. Yang, and T.-S. Yeo, "SAR automatic target recognition based on multiview deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2196–2210, Apr. 2018.

[7] O. Kechagias-Stamatis and N. Aouf, "Fusing deep learning and sparse coding for SAR ATR," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 55, no. 2, pp. 785–797, Apr. 2019.

[8] M. Yang, L. Zhang, X. Feng, and D. Zhang, "Fisher discrimination dictionary learning for sparse representation," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 543–550.

[9] M. Aharon, M. Elad, and A. Bruckstein, "K-SVD: An algorithm for designing overcomplete dictionaries for sparse representation," *IEEE Trans. Signal Process.*, vol. 54, no. 11, pp. 4311–4322, Nov. 2006.

[10] Z. Jiang, Z. Lin, and L. S. Davis, "Label consistent K-SVD: Learning a discriminative dictionary for recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 11, pp. 2651–2664, Nov. 2013.

[11] Y. Huang, G. Liao, Z. Zhang, Y. Xiang, J. Li, and A. Nehorai, "SAR automatic target recognition using joint low-rank and sparse multiview denoising," *IEEE Geosci. Remote Sens. Lett.*, Vol. 15, no. 10, pp. 1570–1574, Jul. 2018.

[12] M. Yu, S. Zhang, L. Zhang, L. Zhao, and G. Kuang, "SAR target recognition via joint manifold regularized low-rank matrix approximation," in *Proc. 19th Int. Radar Symp. (IRS)*, Jun. 2018, pp. 1–7.

[13] I. G. Cumming and F. H. Wong, *Digital Processing of Synthetic Aperture Radar Data: Algorithms and Implementation.* Norwood, MA, USA: Artech House, 2005.

[14] J.-S. Lee, "A simple speckle smoothing algorithm for synthetic aperture radar images," *IEEE Trans. Syst., Man, Cybern.*, vols. SMC–13, no. 1, pp. 85–89, Jan. 1983.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:50:41 UTC from IEEE Xplore. Restrictions apply.

[15] A. M. Raynal, D. L. Bickel, and A. W. Doerry, "Stationary and moving target shadow characteristics in synthetic aperture radar," *Proc. SPIE*, vol. 9077, May 2014, Art. no. 90771B.

[16] D. T. Kuan, A. A. Sawchuk, T. C. Strand, and P. Chavel, "Adaptive noise smoothing filter for images with signal-dependent noise," *IEEE Trans. Pattern Anal. Mach. Intell.*, vols. PAMI–7, no. 2, pp. 165–177, Mar. 1985.

[17] V. S. Frost, J. A. Stiles, K. S. Shanmugan, and J. C. Holtzman, "A model for radar images and its application to adaptive digital filtering of multiplicative noise," *IEEE Trans. Pattern Anal. Mach. Intell.*, vols. PAMI–4, no. 2, pp. 157–166, Mar. 1982.

[18] Y. Yu and S. T. Acton, "Speckle reducing anisotropic diffusion," *IEEE Trans. Image Process.*, vol. 11, no. 11, pp. 1260–1270, Nov. 2002.

[19] S. Parrilli, M. Poderico, C. V. Angelino, and L. Verdoliva, "A nonlocal SAR image denoising algorithm based on LLMMSE wavelet shrinkage," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 2, pp. 606–616, Feb. 2012.

[20] C.-A. Deledalle, L. Denis, and F. Tupin, "Iterative weighted maximum likelihood denoising with probabilistic patch-based weights," *IEEE Trans. Image Process.*, vol. 18, no. 12, pp. 2661–2672, Dec. 2009.

[21] R. Touzi, "A review of speckle filtering in the context of estimation theory," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 11, pp. 2392–2404, Nov. 2002.

[22] F. Argenti, A. Lapini, T. Bianchi, and L. Alparone, "A tutorial on speckle reduction in synthetic aperture radar images," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 3, pp. 6–35, Sep. 2013.

[23] S. Foucher and C. Lopez-Martinez, "Analysis, evaluation, and comparison of polarimetric SAR speckle filtering techniques," *IEEE Trans. Image Process.*, vol. 23, no. 4, pp. 1751–1764, Apr. 2014.

[24] A. Ishimaru, T.-K. Chan, and Y. Kuga, "An imaging technique using confocal circular synthetic aperture radar," *IEEE Trans. Geosci. Remote Sens.*, vol. 36, no. 5, pp. 1524–1530, Dec. 1998.

[25] N. Vaswani, T. Bouwmans, S. Javed, and P. Narayanamurthy, "Robust subspace learning: Robust PCA, robust subspace tracking, and robust subspace recovery," *IEEE Signal Process. Mag.*, vol. 35, no. 4, pp. 32–55, Jul. 2018.

[26] Z. Lin, M. Chen, and L. Wu, "The augmented Lagrange multiplier method for exact recovery of corrupted low-rank matrices," Univ. Illinois Urbana-Champaign, Champaign, IL, USA, Tech. Rep. UILU-ENG-09-2215, 2009.

[27] S. J. Wright, R. D. Nowak, and M. A. T. Figueiredo, "Sparse reconstruction by separable approximation," *IEEE Trans. Signal Process.*, vol. 57, no. 7, pp. 2479–2493, Jul. 2009.

[28] C. Ozcan, B. Sen, and F. Nar, "GPU efficient SAR image despeckling using mixednorms," in *Proc. SPIE*, vol. 9247, Oct. 1014, Art. no. 92470D.

[29] L. M. Kaplan, "Analysis of multiplicative speckle models for template-based SAR ATR," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 37, no. 4, pp. 1424–1432, Jan. 2001.

[30] A. Beck and M. Teboulle, "Fast gradient-based algorithms for constrained total variation image denoising and deblurring problems," *IEEE Trans. Image Process.*, vol. 18, no. 11, pp. 2419–2434, Nov. 2009.

[31] M. J. D. Powell, "A method for nonlinear constraints in minimization problems," in *Optimization*, R. Fletcher, Ed. New York, NY, USA: Academic, 1969, pp. 283–298.

[32] B. Hou, Z. Wen, L. Jiao, and Q. Wu, "Target-oriented high-resolution SAR image formation via semantic information guided regularizations," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1922–1939, Apr. 2018.

[33] J.-F. Cai, E. J. Candś, and Z. Shen, "A singular value thresholding algorithm for matrix completion," *SIAM J. Optim.*, vol. 20, no. 4, pp. 1956–1982, Jan. 2010.

[34] Y. Ma, C. Li, X. Mei, C. Liu, and J. Ma, "Robust sparse hyperspectral unmixing with $\ell_{2,1}$ norm," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1227–1239, Mar. 2017.

[35] S. Boyd, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, 2010.

[36] B. He, M. Tao, M. Xu, and X. Yuan, "Alternating directions based contraction method for generally separable linearly constrained convex programming problems," *SIAM J. Optimiz.*, vol. 62, no. 4, pp. 2479–2493, 2011.

[37] D. L. Donoho, "De-noising by soft-thresholding," *IEEE Trans. Inf. Theory*, vol. 41, no. 3, pp. 613–627, May 1995.

[38] M. Tao and J. Yang, "Alternating direction algorithms for total variation deconvolution in image reconstruction," Dept. Math., Nanjing Univ., Nanjing, China, Tech. Rep. TR0918, 2009.

[39] S. H. Chan, R. Khoshabeh, K. B. Gibson, P. E. Gill, and T. Q. Nguyen, "An augmented lagrangian method for total variation video restoration," *IEEE Trans. Image Process.*, vol. 20, no. 11, pp. 3097–3111, Nov. 2011.

[40] W.-J. Zeng and H. C. So, "Outlier-robust matrix completion via ℓ_p -Minimization," *IEEE Trans. Signal Process.*, vol. 66, no. 5, pp. 1125–1140, Mar. 2018.

[41] T. D. Ross, S. W. Worrell, V. J. Velten, J. C. Mossing, and M. L. Bryant, "Standard SAR ATR evaluation experiments using the MSTAR public release data set," *Proc. SPIE*, vol. 3370, pp. 566–573, Sep. 1998.

[42] G. R. Benitz, "High-definition vector imaging," *Lincoln Lab. J.*, vol. 10, no. 2, pp. 147–170, Mar. 1997.

[43] C. E. Shannon, "A mathematical theory of communication," *Bell Syst. Tech. J.*, vol. 27, no. 3, pp. 379–423, Jul./Oct. 1948.

[44] W.-J. Zeng, H. C. So, and L. Huang, "ℓ_p-MUSIC: Robust direction-of-arrival estimator for impulsive noise environments," *IEEE Trans. Signal Process.*, vol. 61, no. 17, pp. 4296–4308, Jun. 2013.

[45] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[46] A. Achim, P. Tsakalides, and A. Bezerianos, "SAR image denoising via Bayesian wavelet shrinkage based on heavy-tailed modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 8, pp. 1773–1784, Aug. 2003.

[47] Y. Xu, "Fast algorithms for higher-order singular value decomposition from incomplete data," *J. Comput. Math.*, vol. 35, no. 4, pp. 397–422, Jun. 2017.



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